

Supplementary Material

Who verifies the number? Market concentration and verifier effects in mandatory emissions verification

All tables and figures in this document are produced by scripts from the result files of the reproduction package (S12); no number is typed by hand. Verifier identities are coded V1–V24 in descending order of 2013–2024 client-years; V1 is the reference verifier in all fixed-effects specifications. Coefficients on log outcomes are reported as log points $\times 100$. Section numbers follow the references in the main text.

S1. Panel construction and reconciliation

The panel is assembled from the annual key-information workbooks of entity-level emissions statements for 2011–2024 (12,731 rows). Two rows without a verifier are dropped. Verifier names are harmonised to 24 codes (alias table in the reproduction package). Rows with non-positive or missing energy (27) or non-positive emissions (4) are excluded from log outcomes only. The 2012 verifier column duplicates the 2011 column for 99.1% of entities (84.1% agreement between 2012 and 2013, 82.0% between 2013 and 2014), so all series that depend on year-to-year changes start in 2013. The designated-industry field is bridged from 2020: each entity’s last designated industry is carried forward; new entities receive the industry implied by a crosswalk from the planning industry (2019–2021 pairs); 69 entity-years use the planning industry directly and 11 keep a coarse sector label. Table S1 gives the counts by year.

Of the 1,848 entities in the panel, 1,827 are observed from 2013 (21 are observed only in 2011–2012). Of those 1,827, 613 are observed with at least two verifiers (movers) and 1,214 with one. In Table S2 the number of never-movers differs across rows because the ETS share excludes 14 entities whose ETS status is missing in every year, and the within-entity standard deviation of $\ln$ intensity requires at least two valid observations (1,632 entities: 613 movers and 1,019 never-movers).

Table S1: Panel construction by registry year (industry label source: designated in the workbook, carried forward from the entity’s last designated label, crosswalk from the planning industry, or other = planning industry used directly or coarse sector label)

Year	Entity-yrs	Verif.	Listed	ETS	Desig.	Carried	Crossw.	Other	Cov.	pre-bridge	Excl. logs
2011	450	23	116		450	0	0	0		1.00	0
2012	547	23	139		547	0	0	0		1.00	0
2013	599	21	147		599	0	0	0		1.00	0
2014	855	19	188		855	0	0	0		1.00	0
2015	853	19	189	0.71	853	0	0	0		1.00	0
2016	895	19	196	0.67	895	0	0	0		1.00	0
2017	956	18	208	0.62	956	0	0	0		1.00	0
2018	1002	17	212	0.58	994	2	0	6		0.99	0
2019	1043	16	219	0.59	1041	1	0	1		1.00	1

Year	Entity-yrs	Verif.	Listed	ETS	Desig.	Carried	Crossw.	Other	Cov. pre-bridge	Excl. logs
2020	1058	16	232	0.58	935	93	26	4	0.88	4
2021	1077	15	241	0.64	569	463	37	8	0.53	2
2022	1094	13	270	0.64	388	622	66	18	0.35	3
2023	1136	13	283	0.64	404	626	85	21	0.36	7
2024	1164	13	289	0.67	389	650	103	22	0.33	10

S2. Movers versus never-movers

Table S2: Movers versus never-movers, 2013–2024

Variable	Movers: mean	median	Never-movers: mean	median	<i>n</i> movers	<i>n</i> never-movers
years observed	8.896	11.000	5.172	4.000	613	1214
median emissions (tCO ₂)	770,064	56,005	240,659	33,180	613	1214
ETS share	0.551	0.778	0.387	0.000	613	1200
within-entity SD of ln intensity	0.106	0.020	0.061	0.012	613	1019
waste or industrial-complex industry	0.142	0.000	0.071	0.000	613	1214
listed company	0.212	0.000	0.161	0.000	613	1214

S3. Industry split and Annex 5

Annex 5 of the MRV guideline (Ministry of Environment Notice 2025-64) sets the minimum calculation tier by emission activity and facility size: group A below 50 ktCO₂, group B from 50 to 500 kt, group C at 500 kt and above. For stationary combustion the minimum tier is 1, 2 and 3 in groups A, B and C; for mobile combustion 1, 1 and 2 (1 for rail and ships); for purchased electricity and heat the national factor applies at every size. Process activities (minerals, chemicals, metals, electronics) and waste treatment allow plant-specific parameters in group B or C or use activity-data tiers that vary by process. Each of the 77 designated-industry labels was assigned to an activity class by its dominant emission activity; the fixed-factor group comprises entity-years in the combustion, mobile, building and energy classes with emissions below 500 kt, so that no facility can be in group C. The Target Management Scheme applies the same annex by reference (Notice 2020-3, Article 84(1)). Table S3 summarises the split; the label-level table with the rationale for each label is in the reproduction package (industry_split.csv). Only the current version of the annex was obtained; earlier versions are assumed to have the same tier structure.

Table S3: Industry split by activity class (77 designated-industry labels)

Activity class	Group (below 500 kt)	Industries	Entity-years	Entities	Median residual SD
building	fixed-factor	8	1236	192	0.056
combustion	fixed-factor	28	3303	581	0.052
energy	fixed-factor	4	541	113	0.661
mobile	fixed-factor	14	944	227	0.114
process	discretion	17	4551	807	0.297
waste	discretion	6	1130	240	0.946

S4. Institutional chronology and NIER designation records

Table S4: Chronology of the instruments used in the paper

Date	Event and source
2011	First designations of verification bodies under the Target Management Scheme (designation numbers 2011-1 to 2011-25; NIER register).
2015-01	K-ETS starts; verified statements of ETS entities enter the same registry.
2017-05-30	Verification guideline (Notice 2017-12) sets the 3-year designation validity (Article 21(6)); the proviso resetting the clock for earlier designations expired on 2017-06-30.
2018-02-28	Cancellation of the designation of one verifier (V19); cause not recorded (NIER register, 2024-07-01 issue).
2019-02-07	Cancellation, V18; cause not recorded.
2019-08-22	Ministry of Environment publishes per-verifier error rates, contribution rates and correction rates for 17 verifiers (2018 statements; 3,586 errors found in the conformity assessment).
2019-11-01	Cancellation, V16; cause not recorded.
2020-10-27	Ministry of Environment publishes a composite 100-point score (appropriateness 50, scale 10, operation 40) for 16 verifiers, with an announced annual cycle; no later public round was found (searches of the ministry and NGMS notice boards, 2026-09-19).
2021-06-08	Verification guideline (Notice 2021-112) delegates designation, management and sanction powers to NIER.
2021-09-16	NIER Notice 2021-65 creates a separate ISO/IEC 17029 and ISO 14065 accreditation track for international mutual recognition, operated separately from statutory designation; no transition deadline.
2022-02	NIER joins the IAF multilateral recognition arrangement for greenhouse-gas verification.
2022-03-01	Cancellation, V17; cause not recorded.
2022-06-16 to 06-30	First accreditation applications (15 verifiers).
2022-08-01 to 2023-01-31	Suspension of V15; cancellation 2022-10-01; cause not recorded.
2022-08-08 to 10-24	First 13 accreditations issued.
2022-09-02	Cancellation, V4; cause not recorded.
2023-08 to 09	Re-designation of all 13 active verifiers, valid to 2026.

Sources: law.go.kr texts of Notices 2017-12, 2018-70, 2021-112 and 2024-169 (verification guideline), Notice 2025-64 (MRV guideline), NIER Notice 2021-65; NIER register of designations, suspensions and cancellations (2024-07-01) and designation status (2025-01-06); Ministry of Environment press releases of 2019-08-22 and 2020-10-27; URLs are listed in the reproduction package (external/collection_status.md). The verification guideline contains no mandatory rotation of verification bodies or auditors and no cap on consecutive years with the same client (checked in the 2018, 2021 and 2024 versions).

Table S5: Verifier roster: registry presence and NIER designation events (cancellation cause is not recorded in the register)

Code	First	Last	Peak clients	Clients 2024	Share 2024	Status	Cancelled	Suspended
V8	2011	2024	35	34	0.029	active		
V9	2011	2024	32	32	0.027	active		2012.9.7 2012.9.23
V6	2011	2024	63	63	0.054	active		
V2	2011	2024	185	170	0.146	active		2021.4.1 2021.4.30
V11	2011	2024	21	18	0.015	active		
V3	2011	2024	266	266	0.229	active		
V10	2011	2024	30	15	0.013	active		
V5	2011	2024	93	93	0.080	active		2020.8.28 2020.9.27
V1	2011	2024	362	346	0.297	active		

Code	First	Last	Peak clients	Clients 2024	Share 2024	Status	Cancelled	Suspended
V12	2011	2024	24	23	0.020	active		2011.10.12 2012.6.30 2013.1.8
V7	2011	2024	70	70	0.060	active		
V13	2011	2024	23	20	0.017	active		
V14	2011	2024	15	14	0.012	active		
V19	2011	2016	18	0	0.000	cancelled	2018.2.28	
V4	2011	2021	97	0	0.000	cancelled	2022.9.2	
V18	2011	2017	17	0	0.000	cancelled	2019.2.7	
V15	2011	2021	19	0	0.000	cancelled	2022.10.1	2022.8.1 2023.1.31
V17	2011	2020	15	0	0.000	cancelled	2022.3.1	
V16	2011	2018	16	0	0.000	cancelled	2019.11.1	
V20	2011	2013	1	0	0.000	pilot dropout	2015.3.13	2012.12.17 2013.12.17 2014.12.16
V23	2011	2011	2	0	0.000	pilot dropout	2015.3.13	2012.6.28 2012.12.1 2013.12.31; 2014.1.1 2014.12.31
V21	2011	2013	5	0	0.000	pilot dropout	2014.12.17	
V24	2011	2012	10	0	0.000	pilot dropout	2015.3.13	2013.11.14 2014.11.13
V22	2012	2012	1	0	0.000	pilot dropout		

Table S6: NIER register: suspensions of designation (all entries)

Code	From	To
V12	2011.10.12	2011.1.29
not in registry	2012.6.28	2012.12.14
V23	2012.6.28	2012.11.28
V12	2012.6.30	2013.1.8
V9	2012.9.7	2012.9.23
V23	2012.12.1	2013.12.31
V20	2012.12.17	2013.12.16
V24	2013.11.14	2014.11.13
not in registry	2013.12.15	2014.12.14
V20	2013.12.17	2014.12.16
V23	2014.1.1	2014.12.31
V5	2020.8.28	2020.9.27
V2	2021.4.1	2021.4.30
V15	2022.8.1	2023.1.31
V2	2024.1.1	2024.6.30

Table S7 gives the full text of Table 1 of the main paper, with the legal instrument and article from which each cell was read. The EU cells were read from the consolidated texts of Regulation (EU) 2018/2067 (Accreditation and Verification), Regulation (EU) 2018/2066 (Monitoring and Reporting), Directive 2003/87/EC and Regulation (EC) 765/2008 on EUR-Lex, and from the Commission’s Explanatory Guidance Document 1, its quick guide for verifiers, its FAQ on accreditation and verification, and the Commission staff working document accompanying the 2021 carbon market report. The China cells were read from the Interim Regulations on Carbon Emission Trading (State Council Order 775, 2024) and the Administrative Measures (MEE Order 19, 2021) on gov.cn and mee.gov.cn, from the Verification Guideline (MEE 2021-130), the power-sector accounting and reporting guideline (MEE 2022-485), the annual MEE work notices of 2022 to 2026, the MEE explanatory note of 15 March 2022, the 2025 Opinion on the national carbon market, and the National Carbon Market Reports of 2024 and 2025. URLs are in the reproduction package (external/eu_china_governance.csv).

Table S7: Verification governance instruments in Korea, the EU and China: full text of Table 1 with the instrument each cell was read from

Who selects and pays the verifier

Korea. The verified entity contracts and pays the verifier; an impartiality self-assessment (Annex Form 1) is attached to every contract (verification guideline Art. 13(2))

Source: <https://www.law.go.kr/LSW//admRulInfoP.do?admRulSeq=2100000246354&chrClsCd=010201>

EU ETS. The operator contracts and pays the verifier; Commission guidance treats the operator's payment as a source-of-revenue risk to impartiality.

Source: AVR Art. 7(3), 8(1); Explanatory Guidance Document 1, sec. 5.2

China. The provincial ecology and environment authority is the verifying body and may entrust technical service institutions through government purchase of services; the emitter contracts institutions only for testing and for preparing the report. Provinces budget for the next year's verification bodies and complete the tender before year-end.

Source: Interim Regulations on Carbon Emission Trading (State Council Order 775, 2024) Art. 11-12; Administrative Measures (MEE Order 19, 2021) Art. 26; MEE notice 2022-111 sec. 3(3)

Verifier independence rules

Korea. Entity level: no consulting or abatement-service firm may be designated (Art. 21(4)6); no advisory services to the same client and a 2-year cooling-off for auditors who advised it (Art. 24(5)). Person level: 2-year cooling-off for former employees and consultants (Art. 9(4))

Source: <https://www.law.go.kr/LSW//admRulInfoP.do?admRulSeq=2100000246354&chrClsCd=010201>

EU ETS. The verifier must be independent of the operator and of the competent authority. Consulting or technical assistance on monitoring and reporting for the same client is an unacceptable impartiality risk; common ownership, governance, management, personnel, resources or finances create a conflict of interest; a documented process must maintain continuous impartiality; and when verifying the same operator as in the previous year the verifier must consider and mitigate the impartiality risk. Guidance suggests at least two years before staff with a consulting history join a verification.

Source: AVR Art. 43(1)-(6a), Annex II; Explanatory Guidance Document 1, sec. 5.2

China. Institutions may not provide allowance calculation, consulting or management services to the emitter, engage in carbon-asset management or trading, hold asset or management ties with it, share managers with it or have mutually employed its managers within three years, or use verifiers employed by or having advised the emitter within three years. Within one province an institution may not both prepare annual reports and conduct technical review.

Source: Verification Guideline (MEE 2021-130) sec. 4.1.1; Interim Regulations Art. 13

Mandatory verifier rotation

Korea. None, at either the body or the auditor level; no cap on consecutive years with the same client (checked in Notices 2018-70, 2021-112 and 2024-169)

Source: <https://www.law.go.kr/LSW//admRulInfoP.do?admRulSeq=2100000246354&chrClsCd=010201>

EU ETS. No rotation of the verification body, but mandatory rotation of the EU ETS lead auditor: after five consecutive annual verifications of the same installation (six for an aircraft operator) the lead auditor must take a three-year break. The five years count from 2021 data, so the first rotation follows the verification of 2025 data.

Source: AVR Art. 43(7)-(8), inserted by Reg. (EU) 2020/2084; Explanatory Guidance Document 1, sec. 5.3

China. None found in the Interim Regulations, the Administrative Measures, the Verification Guideline or the annual MEE work notices. The guideline instead allows the on-site visit to be waived for emitters with no non-conformity in the two preceding years.

Source: Verification Guideline sec. 4.1.4 (searched: Interim Regulations, Measures, Guideline, MEE notices 2022-2026)

Designation or accreditation

Korea. Statutory designation by NIER since 2021-06-08 (before that, the Ministry of Environment); 3-year validity with re-designation filing (Art. 21(6)-(7)) and an annual NIER assessment of each verifier (Art. 23(1)). A separate ISO/IEC 17029 and ISO 14065 accreditation track was created by NIER Notice 2021-65 (2021-09-16) for international mutual recognition; the first 13 accreditations were issued 2022-08-08 to 2022-10-24

Source: <https://www.law.go.kr/LSW//admRulLsInfoP.do?admRulSeq=2100000204634>

EU ETS. Accreditation for the scope of Annex I by the single national accreditation body of each Member State designated under Regulation 765/2008, against the harmonised standards referred to in Annexes II-III (EN ISO 14065 and ISO/IEC 17029 in national practice). The certificate is valid for at most five years, with annual surveillance, reassessment before expiry, extraordinary assessments, and peer evaluation of the accreditation bodies themselves.

Source: AVR Art. 44, 49(2), 50-52, 55, 65; Reg. (EC) 765/2008 Art. 4(1), 5(1)

China. No qualification approval or accreditation for verification institutions; provincial authorities select them by tender under government purchase of services. Institutions must have appropriate facilities, technical capacity, personnel and a quality management system, with detailed measures still to be issued. A 2025 Party Central Committee and State Council opinion directs the introduction of certification-body qualification management with entry conditions, conduct norms and an exit mechanism.

Source: Interim Regulations Art. 13; MEE explanatory note of 15 March 2022; Opinion on the national carbon market (25 Aug. 2025) sec. 13

Minimum tier rules that fix factors

Korea. Annex 5 of the MRV guideline: minimum tier by activity and facility size (A below 50 kt: Tier 1; B 50-500 kt: Tier 2; C at or above 500 kt: Tier 3 for combustion); indirect emissions from purchased electricity and heat use the national factor at every size. The Target Management Scheme applies the same annex by reference (Notice 2020-3, Art. 84(1))

Source: <https://www.law.go.kr/LSW/fDownload.do?fSeq=151023935&bylClsCd=200201>

EU ETS. Installation categories A (at most 50 kt), B (50-500 kt) and C (above 500 kt), with minor and minimis source streams. Category A installations and commercial standard fuels apply the minimum tiers of Annex V; all others must apply the highest tier of Annex II (measured factors), relaxable by one level for category C and two for A and B where the required tier is technically infeasible or unreasonably costly.

Source: MRR Art. 19(2)-(3), 26(1); Annexes II and V

China. No size-based tier structure. In the power sector the guideline requires sampling and testing of the carbon content of coal by a CMA-certified or CNAS-accredited laboratory within 40 days of sampling; where testing is absent or non-compliant, a conservative default of 0.03085 tC per GJ applies (0.02858 for unconventional coal units).

Source: Power Sector Accounting and Reporting Guideline (MEE 2022-485) sec. 6.2.2, 6.2.4.1; MEE notice 2022-111 sec. 4

Regulator's own conformity assessment of certified values

Korea. KECO conformity assessment after verification (3,586 errors found in the 2018 statements according to the 2019-08-22 release); the certified values in the registry are post-assessment

Source: <https://www.korea.kr/news/policyNewsView.do?newsId=156346869>

EU ETS. The competent authority makes a conservative estimate of emissions where no verified report is submitted, where the report does not comply with the MRR, or where verification did not follow the AVR, and assesses uncorrected non-material misstatements. It communicates to the accreditation body each year the verifier non-compliance found in those checks, which the accreditation body must treat as a complaint and answer within three months.

Source: MRR Art. 68(1), 70(1)-(2); AVR Art. 73(1)-(2); Directive 2003/87/EC Art. 15

China. Verification is itself the provincial authority's administrative act: results are fed back within seven working days and published, and an emitter may seek review within seven working days. MEE runs a national platform for whole-process supervision and both levels may inspect institutions on site. Monthly archived data pass a national-provincial-municipal joint review (over three million power-sector records in 2024) and MEE has run three rounds of on-site supervision covering 538 emitters since 2021.

Source: Interim Regulations Art. 12, 16-17; Administrative Measures Art. 27, 31; National Carbon Market Report 2024

Public evaluation of verifiers

Korea. Two rounds only: 2019-08-22 (17 verifiers; error rate, contribution rate and correction rate per verifier, ranked) and 2020-10-27 (16 verifiers; composite 100-point score). The guideline provides for an annual NIER assessment, but no public round was found for 2021-2024 despite the announced annual cycle

Source: <https://www.mcee.go.kr/home/web/board/read.do?menuId=&boardMasterId=1&boardCategoryId=39&boardId=1405640>

EU ETS. No published performance ranking of verifiers. Accreditation bodies must keep a public database of accredited verifiers, their scope and any administrative measures imposed on them; their management reports on surveillance and complaints go to the competent authority, not to the public. The Commission publishes only country-level counts of complaints and measures.

Source: AVR Art. 71(3), 76(1)-(2); Commission SWD(2021) 308 sec. 11.2

China. Annual and mandatory: after each verification round the provincial authority evaluates the quality, compliance and timeliness of every technical service institution and publishes the result on its website in a prescribed disclosure form, together with every verification conclusion. In 2023 the provinces published 16,600 evaluation records, 99.7 per cent of them qualified. In March 2022 MEE named four institutions for falsified or perfunctory verification.

Source: Verification Guideline sec. 6 and Annex 6; MEE notices 2022-111, 2023-43, 2025-140, 2026-32; National Carbon Market Report 2024; MEE disclosure of 14 March 2022

Suspension and cancellation of designation

Korea. The NIER register lists suspensions and cancellations by date; 6 verifiers active in the registry were cancelled between 2018 and 2022, with the cause column blank for all of them; the pilot-period dropouts were cancelled in 2014-2015

Source: <https://nier.go.kr/comm/downloadFile.do?url=aa40d64c-18a9-4f7f-858e-97d85d65689a>

EU ETS. The accreditation body must suspend or restrict accreditation for a serious or persistent breach and must withdraw it where the grounds for suspension are not remedied, where management or staff are found guilty of fraud, or where false information was intentionally supplied. Measures take effect on notification, are communicated to the competent authority and accreditation body of every Member State in which the verifier works, and appear in the public database. For 2020 the Member States reported no suspensions, one withdrawal and two reductions of scope.

Source: AVR Art. 54(1)-(5), 62, 72, 76(2)(e); Commission SWD(2021) 308 Table 11.2.1

China. There is no accreditation to suspend. A report or technical review with major defects, or tampered or forged data, draws confiscation of gains and a fine of 5 to 10 times those gains, or RMB 200,000 to 1,000,000 where there are none, and in serious cases a ban from the business; responsible individuals face five-year or lifetime bans, and penalties enter the published credit record. After the March 2022 disclosure MEE instructed localities to be prudent in entrusting the named institutions.

Source: Interim Regulations Art. 23, 27; MEE notice 2022-111 sec. 3(1)

Procedure for client books after a cancellation

Korea. None found in public records; the observed pattern is absorption by a single successor (V4 to V3 87%, V17 to V3 100%, V18 to V1 81% of the cancelled verifier's last-year clients)

Source: [external/exit_classification.csv](#); [results/transition_matrix.csv](#)

EU ETS. The operator must hire another verifier to have the report verified; the new verifier may use the previous verifier's information only so far as its own risk analysis allows, and operators are advised to provide for disclosure in their contracts. The AVR contains no transitional rule. If no AVR-compliant verification is in place by 31 March, the competent authority estimates emissions conservatively and allowance transfers are blocked.

Source: Commission FAQ on accreditation and verification (21 March 2022) Q1; MRR Art. 70(1)(c); Directive Art. 15

China. No explicit rule. Because the provincial authority procures the institutions and allocates verification tasks among them, reassignment falls to it.

Source: Verification Guideline sec. 4.1.1-4.1.2; MEE notice 2022-111 sec. 2(2) (searched: Interim Regulations, Measures, Guideline, MEE notices 2022-2026)

S5. Switching and entrants' choices

Table S8: Switch decomposition by year (consecutive pairs, 2013 onwards)

Year	Pairs	Switches	Voluntary	Block	Dispersed	Pilot	Rate	Scope change: switches	stayers
2013	532	85	74	0	0	11	0.160	0.035	0.047
2014	568	103	102	0	0	1	0.181	0.155	0.170
2015	840	120	120	0	0	0	0.143	0.142	0.217
2016	822	113	113	0	0	0	0.137	0.062	0.041
2017	807	100	85	0	15	0	0.124	0.110	0.051
2018	902	105	89	13	3	0	0.116	0.152	0.117
2019	924	81	70	7	4	0	0.088	0.148	0.215
2020	962	55	55	0	0	0	0.057	0.073	0.058
2021	1011	81	67	14	0	0	0.080	0.148	0.105

Year	Pairs	Switches	Voluntary	Block	Dispersed	Pilot	Rate	Scope change: switches	stayers
2022	906	112	22	76	14	0	0.124	0.304	0.191
2023	1035	79	79	0	0	0	0.076	0.101	0.053
2024	1080	68	68	0	0	0	0.063	0.176	0.070

Table S9: Annual switching rates by group

Dimension	Category	Pairs	Switch rate	Voluntary rate
emis_decile	1	1039	0.078	0.061
emis_decile	2	1039	0.080	0.067
emis_decile	3	1039	0.101	0.092
emis_decile	4	1039	0.103	0.080
emis_decile	5	1039	0.116	0.092
emis_decile	6	1038	0.091	0.068
emis_decile	7	1039	0.117	0.100
emis_decile	8	1039	0.103	0.092
emis_decile	9	1039	0.118	0.109
emis_decile	10	1039	0.153	0.146
designation	site	5964	0.099	0.080
designation	company	4425	0.116	0.106
ministry	MOLIT	1277	0.132	0.114
ministry	MAFRA	445	0.081	0.074
ministry	MOTIE (2025 name)	176	0.051	0.051
ministry	MOTIE	4509	0.107	0.096
ministry	MOF	64	0.031	0.031
ministry	MOE	541	0.240	0.229
ets_status	ETS	5032	0.093	0.071
ets_status	TM	2595	0.082	0.068
listed	0	7915	0.112	0.096
listed	1	2474	0.086	0.074

Table S10: Entrants' first verifier choice versus incumbents, by year

Year	Entrants	HHI entrants	Top-3 share entrants	HHI incumbents	Top-3 share incumbents
2014	287	1923	0.578	1241	0.491
2016	66	3062	0.758	1414	0.536
2017	141	1750	0.645	1552	0.568
2018	95	2066	0.674	1650	0.594
2019	116	2500	0.759	1635	0.594
2020	91	2293	0.703	1659	0.597
2021	62	1941	0.661	1732	0.626
2022	179	2051	0.749	1945	0.697
2023	99	2093	0.707	1761	0.674
2024	81	2836	0.815	1720	0.661

Table S11: Entrants' first verifier choice versus incumbents, by group (2014–2024 pooled)

Dimension	Category	Entrants	Top-3: entr.	Incumb.	Top-3: inc.	Gap	HHI entr.	HHI inc.
regime	ETS	263	0.646	6120	0.582	0.065	1806	1464
regime	TM	661	0.743	3048	0.682	0.060	2207	2039
regime	pre-ETS(2013-14)	287	0.578	568	0.491	0.087	1923	1241
regime	unknown	17	0.588	169	0.497	0.091	2180	1314

Dimension	Category	Entrants	Top-3: entr.	Incumb.	Top-3: inc.	Gap	HHI entr.	HHI inc.
ministry	MOLIT	219	0.790	1216	0.666	0.124	2953	1998
ministry	MAFRA	43	0.698	413	0.736	-0.038	2353	2245
ministry	MOTIE	646	0.676	4314	0.596	0.081	2117	1724
ministry	MOF	27	0.630	64	0.531	0.098	2538	2954
ministry	MOE	147	0.558	534	0.395	0.163	1731	1522
designation	site	911	0.696	5826	0.619	0.076	2110	1676
designation	company	313	0.642	4079	0.587	0.056	1776	1547
emis_decile	1	188	0.622	931	0.625	-0.003	2130	1826
emis_decile	2	227	0.736	885	0.677	0.059	2263	2071
emis_decile	3	165	0.745	946	0.659	0.087	2444	2063
emis_decile	4	139	0.669	974	0.588	0.081	2068	1608
emis_decile	5	122	0.713	992	0.595	0.118	1870	1541
emis_decile	6	119	0.664	992	0.590	0.074	2013	1464
emis_decile	7	74	0.635	1037	0.595	0.040	1837	1418
emis_decile	8	74	0.649	1039	0.594	0.055	1790	1544
emis_decile	9	71	0.662	1039	0.541	0.121	2089	1649
emis_decile	10	49	0.592	1070	0.611	-0.019	1912	1770

Table S12: Net flows into the 2024 top-3 by switch type and period

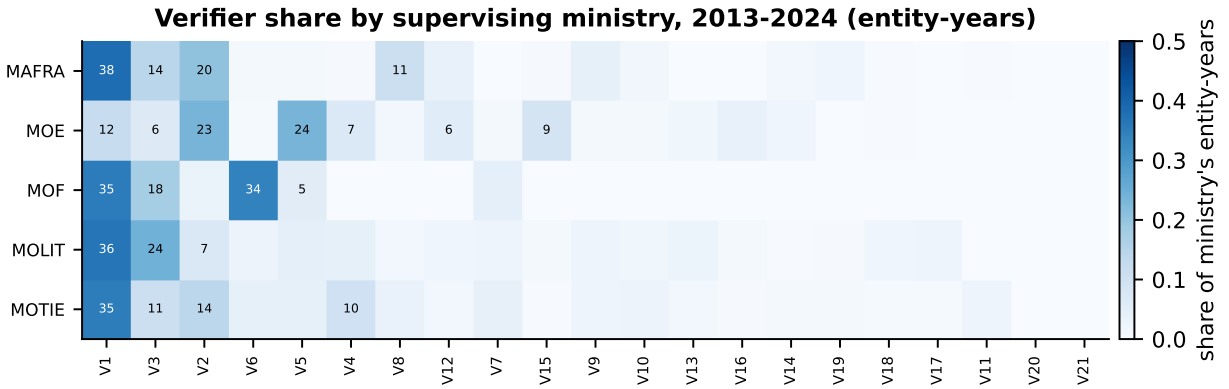
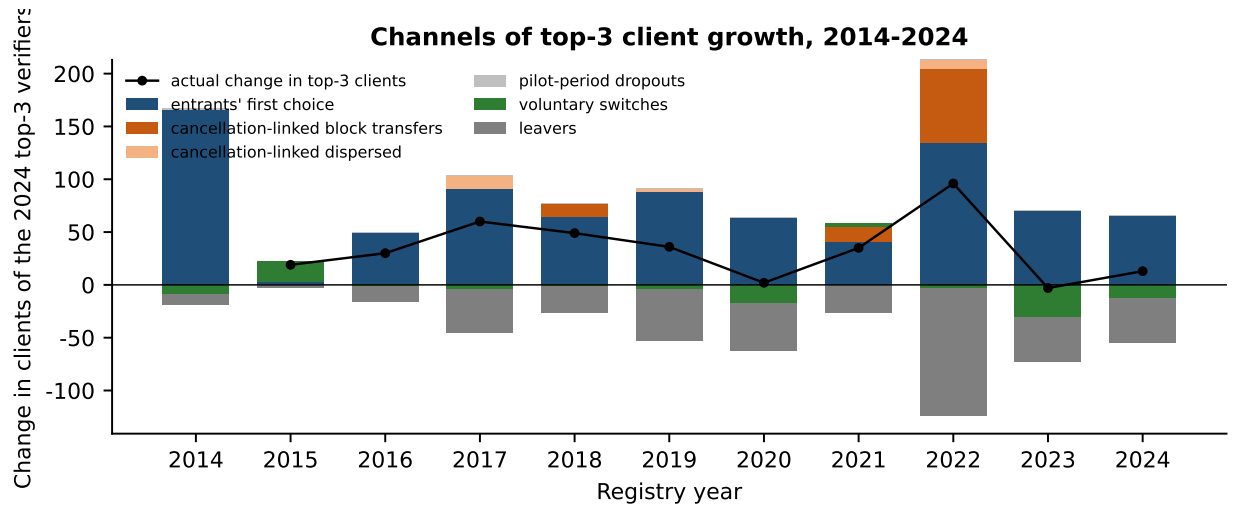
Type	Period	Net into top-3	Switches
exit_block	pairs_2014plus	98	110
exit_dispersed	pairs_2014plus	24	36
pilot	pairs_2013(2012->2013)	2	11
pilot	pairs_2014plus	1	1
voluntary	pairs_2013(2012->2013)	17	74
voluntary	pairs_2014plus	-61	870

Table S13: Decomposition of the fixed top-3 share change, 2013–2024

Term	Value
share_2013	0.5008
share_2024	0.6718
delta_share	0.1710
numerator_entrants_top3	0.7191
numerator_leavers_top3	-0.3600
numerator_net_voluntary	-0.0524
numerator_net_exit_block	0.0842
numerator_net_exit_dispersed	0.0206
numerator_net_pilot	0.0009
numerator_residual_reentry	0.0017
denominator_entrants_total	-0.5284
denominator_leavers_total	0.2853
denominator_residual	0.0000
net_entrants(num+den)	0.1907
net_leavers(num+den)	-0.0747
entrants_excess_top3_clients_vs_incumbent_share	110.9676
share_2024_cf_entrants_at_incumbent_share	0.5765

Table S14: Share of the registry in the fixed-factor group, all entities

Sample	Year	Entity-years	Share of entity-years	Share of emissions	Share of energy
all entities	2013	599	0.409	0.041	0.061
all entities	2014	855	0.451	0.050	0.077
all entities	2015	853	0.447	0.051	0.077
all entities	2016	895	0.451	0.051	0.079
all entities	2017	956	0.454	0.049	0.076
all entities	2018	1002	0.469	0.049	0.073
all entities	2019	1043	0.488	0.053	0.082
all entities	2020	1057	0.490	0.055	0.082
all entities	2021	1077	0.499	0.057	0.087
all entities	2022	1094	0.479	0.061	0.092
all entities	2023	1135	0.483	0.066	0.097
all entities	2024	1162	0.485	0.068	0.098
all entities	2013-2024	11728	0.471	0.054	0.082

**Figure S1:** Verifier share by supervising ministry, 2013–2024 (entity-years).**Figure S2:** Channels of top-3 client growth by year, 2014–2024.

S6. Verifier effects: all samples, outcomes and robustness checks

Tables S15–S20 report every verifier effect with its standard error, 95% confidence interval, MDE and equivalence flags; Table S21 the pre-specified scope-stable subsample; Table S22 the 24 joint tests with Holm adjustment. The robustness set (Tables S23–S27) adds entity-specific linear trends, re-estimates on ‘quiet movers’ (movers none of whose switches coincided with a change in $\ln$ energy of more than 10%), on all entities of the fixed-factor group and on 2013–2019, reports the change in $\ln$ energy at the switch by destination and origin verifier, the industry-by-year cell counts, and the rank correlations between the published evaluation metrics and the verifier effects. The level effects in the frozen specification are not significant after Holm adjustment, disappear with entity trends and among quiet movers, and are largest for the destination verifier with the largest energy jump at switch-in; the 2013–2019 sample, which predates the industry bridge, gives the same null on intensity.

Table S15: Verifier effects on $\ln$ intensity, fixed-factor group (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V12	30	-2.20	1.98	-6.1	+1.7	5.5	0	0	0.306
V10	31	-0.90	1.10	-3.1	+1.3	3.1	0	1	0.429
V6	36	-0.88	1.81	-4.4	+2.7	5.1	0	1	0.666
V8	23	+0.60	1.34	-2.0	+3.2	3.7	0	1	0.635
V7	41	+1.11	0.92	-0.7	+2.9	2.6	0	1	0.234
V16	14	+1.12	0.67	-0.2	+2.4	1.9	0	1	0.088
V18	22	+1.15	0.80	-0.4	+2.7	2.3	0	1	0.152
V13	24	+1.64	1.15	-0.6	+3.9	3.2	0	1	0.188
V15	18	+1.71	1.39	-1.0	+4.4	3.9	0	1	0.259
V17	13	+1.74	1.58	-1.4	+4.8	4.4	0	1	0.320
V3	171	+1.96	1.62	-1.2	+5.1	4.5	0	1	0.308
V5	42	+2.44	2.30	-2.1	+7.0	6.5	0	0	0.356
V9	38	+2.81	1.80	-0.7	+6.3	5.0	0	0	0.137
V2	120	+2.90	2.14	-1.3	+7.1	6.0	0	0	0.203
V4	122	+3.68	1.76	+0.2	+7.1	4.9	0	0	0.029
V19	11	+12.16	15.60	-18.4	+42.7	43.7	0	0	0.809

Table S16: Verifier effects on $\ln$ emissions, fixed-factor group (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V17	13	-16.33	7.67	-31.4	-1.3	21.5	0	0	0.049
V8	23	-11.90	8.10	-27.8	+4.0	22.7	0	0	0.189
V7	41	-7.76	11.23	-29.8	+14.3	31.4	0	0	0.524
V9	38	-7.48	4.15	-15.6	+0.7	11.6	0	0	0.077
V16	14	-6.11	9.35	-24.4	+12.2	26.2	0	0	0.558
V10	31	-5.61	10.57	-26.3	+15.1	29.6	0	0	0.586
V2	120	+0.34	4.47	-8.4	+9.1	12.5	0	0	0.943
V6	36	+0.48	7.99	-15.2	+16.1	22.4	0	0	0.948
V13	24	+0.87	6.60	-12.1	+13.8	18.5	0	0	0.904
V3	171	+1.64	5.12	-8.4	+11.7	14.3	0	0	0.753
V4	122	+3.75	5.08	-6.2	+13.7	14.2	0	0	0.446
V5	42	+5.28	5.72	-5.9	+16.5	16.0	0	0	0.385
V18	22	+8.53	4.34	+0.0	+17.0	12.2	0	0	0.080
V19	11	+9.80	16.57	-22.7	+42.3	46.4	0	0	0.717
V15	18	+14.43	12.82	-10.7	+39.6	35.9	0	0	0.386

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V12	30	+15.94	8.48	-0.7	+32.6	23.8	0	0	0.040

Table S17: Verifier effects on ln energy, fixed-factor group (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V17	13	-18.08	7.37	-32.5	-3.6	20.6	0	0	0.024
V8	23	-12.50	8.56	-29.3	+4.3	24.0	0	0	0.197
V9	38	-10.29	4.40	-18.9	-1.7	12.3	0	0	0.034
V7	41	-8.86	11.28	-31.0	+13.3	31.6	0	0	0.452
V16	14	-7.23	9.37	-25.6	+11.1	26.2	0	0	0.486
V10	31	-4.71	10.52	-25.3	+15.9	29.5	0	0	0.656
V2	120	-2.57	3.92	-10.2	+5.1	11.0	0	0	0.539
V19	11	-2.36	5.50	-13.1	+8.4	15.4	0	0	0.669
V13	24	-0.77	6.55	-13.6	+12.1	18.3	0	0	0.919
V3	171	-0.32	5.03	-10.2	+9.5	14.1	0	0	0.953
V4	122	+0.07	5.14	-10.0	+10.1	14.4	0	0	0.994
V6	36	+1.36	8.25	-14.8	+17.5	23.1	0	0	0.863
V5	42	+2.84	5.70	-8.3	+14.0	16.0	0	0	0.627
V18	22	+7.38	4.63	-1.7	+16.5	13.0	0	0	0.143
V15	18	+12.72	12.02	-10.8	+36.3	33.7	0	0	0.425
V12	30	+18.15	8.48	+1.5	+34.8	23.7	0	0	0.020

Table S18: Verifier effects on ln intensity, discretion group (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V15	45	-33.43	23.13	-78.8	+11.9	64.8	0	0	0.160
V16	14	-28.86	16.72	-61.6	+3.9	46.8	0	0	0.154
V12	55	-14.07	6.95	-27.7	-0.4	19.5	0	0	0.055
V5	95	-11.63	10.28	-31.8	+8.5	28.8	0	0	0.298
V3	136	-11.07	10.57	-31.8	+9.7	29.6	0	0	0.417
V2	188	-10.93	4.33	-19.4	-2.4	12.1	0	0	0.011
V6	59	-7.85	5.33	-18.3	+2.6	14.9	0	0	0.159
V14	33	-6.30	10.81	-27.5	+14.9	30.3	0	0	0.593
V17	7	-6.06	7.25	-20.3	+8.1	20.3	0	0	0.432
V9	66	-5.53	3.94	-13.3	+2.2	11.0	0	0	0.142
V10	40	-4.14	6.19	-16.3	+8.0	17.3	0	0	0.515
V13	46	-2.63	7.11	-16.6	+11.3	19.9	0	0	0.722
V18	19	-1.14	4.43	-9.8	+7.5	12.4	0	0	0.791
V7	80	+0.48	6.53	-12.3	+13.3	18.3	0	0	0.959
V4	157	+3.00	7.89	-12.5	+18.5	22.1	0	0	0.858
V8	28	+6.18	5.62	-4.8	+17.2	15.7	0	0	0.272

Table S19: Verifier effects on ln intensity, all movers (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V15	60	-17.32	16.44	-49.5	+14.9	46.0	0	0	0.405
V16	28	-14.55	10.22	-34.6	+5.5	28.6	0	0	0.222
V12	83	-7.03	4.02	-14.9	+0.8	11.3	0	0	0.072
V11	15	-6.48	4.23	-14.8	+1.8	11.8	0	0	0.151

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V5	133	-4.47	7.26	-18.7	+9.8	20.3	0	0	0.579
V3	281	-3.65	4.25	-12.0	+4.7	11.9	0	0	0.541
V2	298	-3.27	2.44	-8.1	+1.5	6.8	0	0	0.192
V6	88	-3.06	3.12	-9.2	+3.0	8.7	0	0	0.307
V17	20	-1.89	3.58	-8.9	+5.1	10.0	0	0	0.625
V10	66	-1.50	2.62	-6.6	+3.6	7.3	0	0	0.574
V14	48	-1.23	7.44	-15.8	+13.4	20.8	0	0	0.869
V9	97	-1.19	2.35	-5.8	+3.4	6.6	0	0	0.625
V18	36	-0.65	2.12	-4.8	+3.5	5.9	0	1	0.786
V13	66	+1.59	4.52	-7.3	+10.4	12.6	0	0	0.731
V7	114	+1.62	3.79	-5.8	+9.0	10.6	0	0	0.724
V19	20	+2.95	9.15	-15.0	+20.9	25.6	0	0	0.981
V4	263	+3.65	5.10	-6.4	+13.7	14.3	0	0	0.652
V8	47	+4.87	3.07	-1.1	+10.9	8.6	0	0	0.116

Table S20: Verifier effects on ln intensity, industries with residual SD below 0.15 (post hoc) (log points $\times 100$; reference V1)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V6	31	-0.36	0.79	-1.9	+1.2	2.2	1	1	
V5	35	+0.07	0.56	-1.0	+1.2	1.6	1	1	
V10	20	+0.31	0.54	-0.7	+1.4	1.5	1	1	
V3	138	+0.38	0.60	-0.8	+1.6	1.7	1	1	
V13	27	+0.43	0.48	-0.5	+1.4	1.3	1	1	
V12	21	+0.52	1.01	-1.4	+2.5	2.8	0	1	
V9	30	+0.66	0.56	-0.4	+1.8	1.6	1	1	
V2	98	+0.72	0.41	-0.1	+1.5	1.1	1	1	
V7	46	+0.75	0.64	-0.5	+2.0	1.8	1	1	
V18	18	+0.99	0.67	-0.3	+2.3	1.9	0	1	
V17	15	+1.03	1.30	-1.5	+3.6	3.6	0	1	
V4	87	+1.77	0.90	+0.0	+3.5	2.5	0	1	

Table S21: Verifier effects on ln intensity, fixed-factor group, scope-stable pairs (pre-specified subsample; bootstrap p added post-freeze)

Verifier	Switches	Effect	s.e.	95\% low	95\% high	MDE	TOST 2\%	TOST 5\%	Boot. p
V10	29	-3.43	1.58	-6.5	-0.3	4.4	0	0	0.028
V12	29	-3.09	1.77	-6.6	+0.4	5.0	0	0	0.129
V6	35	-2.89	1.31	-5.5	-0.3	3.7	0	0	0.028
V5	41	-0.47	1.17	-2.8	+1.8	3.3	0	1	0.691
V15	18	-0.46	1.44	-3.3	+2.4	4.0	0	1	0.761
V2	118	-0.02	0.85	-1.7	+1.6	2.4	1	1	0.974
V8	23	+0.13	1.38	-2.6	+2.8	3.9	0	1	0.924
V3	162	+0.42	0.81	-1.2	+2.0	2.3	1	1	0.578
V17	13	+0.70	1.37	-2.0	+3.4	3.8	0	1	0.604
V13	24	+0.74	0.60	-0.4	+1.9	1.7	1	1	0.190
V7	40	+0.85	1.07	-1.2	+3.0	3.0	0	1	0.427
V9	36	+1.33	1.00	-0.6	+3.3	2.8	0	1	0.200
V4	117	+2.05	1.15	-0.2	+4.3	3.2	0	1	0.074

Table S22: The 24 joint tests with Holm adjustment

Sample	Outcome	k	n	Wald	p	Boot. p	Holm p	Holm boot. p	Resid. SD
all movers	ln_int	18	5449	18.050	0.452	0.598	1.000	1.000	0.264
all movers	ln_emis	18	5449	23.007	0.190	0.246	1.000	1.000	0.352
all movers	ln_energy	18	5449	33.308	0.015	0.020	0.322	0.482	0.335
fixed-factor	ln_int	16	2441	17.339	0.364	0.367	1.000	1.000	0.076
fixed-factor	ln_emis	16	2441	34.129	0.005	0.025	0.120	0.578	0.259
fixed-factor	ln_energy	16	2441	36.915	0.002	0.035	0.052	0.739	0.264
discretion	ln_int	16	2916	22.206	0.137	0.241	1.000	1.000	0.346
discretion	ln_emis	16	2916	16.307	0.432	0.608	1.000	1.000	0.367
discretion	ln_energy	16	2916	19.644	0.237	0.407	1.000	1.000	0.325
resid. SD < 0.15	ln_int	12	1624	11.533	0.484	0.608	1.000	1.000	0.038
resid. SD < 0.15	ln_emis	12	1624	14.530	0.268	0.357	1.000	1.000	0.198
resid. SD < 0.15	ln_energy	12	1624	15.874	0.197	0.281	1.000	1.000	0.203
resid. SD $\geq$ 0.15	ln_int	18	3712	18.171	0.444	0.693	1.000	1.000	0.318
resid. SD $\geq$ 0.15	ln_emis	18	3712	25.818	0.104	0.226	1.000	1.000	0.401
resid. SD $\geq$ 0.15	ln_energy	18	3712	31.784	0.023	0.116	0.466	1.000	0.377
scope-stable, all movers	ln_int	18	4398	17.349	0.499	0.593	1.000	1.000	0.212
scope-stable, all movers	ln_emis	18	4398	25.401	0.114	0.246	1.000	1.000	0.308
scope-stable, all movers	ln_energy	18	4398	29.878	0.039	0.111	0.734	1.000	0.299
scope-stable, fixed-factor	ln_int	13	1913	15.366	0.285	0.417	1.000	1.000	0.063
scope-stable, fixed-factor	ln_emis	13	1913	22.951	0.042	0.101	0.761	1.000	0.250
scope-stable, fixed-factor	ln_energy	13	1913	27.494	0.011	0.030	0.234	0.663	0.257
scope-stable, discretion	ln_int	15	2313	20.718	0.146	0.357	1.000	1.000	0.275
scope-stable, discretion	ln_emis	15	2313	16.948	0.322	0.513	1.000	1.000	0.333
scope-stable, discretion	ln_energy	15	2313	20.290	0.161	0.312	1.000	1.000	0.303

Table S23: Robustness set: joint tests, fixed-factor group

Check	Outcome	k	Wald	p	Boot. p	Resid. SD	n	Entities
b_entity_trends/fixed_factor	ln_int	16	16.892	0.393	0.482	0.052	2441	304
b_entity_trends/fixed_factor	ln_emis	16	19.505	0.243	0.322	0.181	2441	304
b_entity_trends/fixed_factor	ln_energy	16	18.438	0.299	0.392	0.187	2441	304
d_quiet_movers/fixed_factor	ln_int	12	6.145	0.909	0.930	0.065	1561	205
d_quiet_movers/fixed_factor	ln_emis	12	13.468	0.336	0.643	0.205	1561	205
d_quiet_movers/fixed_factor	ln_energy	12	11.180	0.514	0.819	0.211	1561	205
e_all_entities/fixed_factor	ln_int	18	25.768	0.105	0.085	0.081	5501	918
e_all_entities/fixed_factor	ln_emis	18	23.678	0.166	0.407	0.256	5501	918
e_all_entities/fixed_factor	ln_energy	18	25.119	0.122	0.352	0.257	5501	918
f_years_2013.2019/fixed_factor	ln_int	13	16.328	0.232	0.191	0.062	1283	260
f_years_2013.2019/fixed_factor	ln_emis	13	19.090	0.120	0.261	0.153	1283	260
f_years_2013.2019/fixed_factor	ln_energy	13	19.928	0.097	0.216	0.157	1283	260
i_scope_stable_boot/fixed_factor	ln_int	13	15.366	0.285	0.317	0.063	1913	294
i_scope_stable_boot/fixed_factor	ln_emis	13	22.951	0.042	0.106	0.250	1913	294
i_scope_stable_boot/fixed_factor	ln_energy	13	27.494	0.011	0.035	0.257	1913	294

Table S24: Robustness set: selected verifier effects (three small verifiers with the largest level effects and three large verifiers)

Check	Outcome	Verifier	Switches	Effect	95\% low	95\% high
b_entity_trends/fixed_factor	ln_int	V9	38	-1.5	-4.4	+1.3
b_entity_trends/fixed_factor	ln_int	V4	122	-1.1	-3.6	+1.4
b_entity_trends/fixed_factor	ln_int	V2	120	-2.0	-4.6	+0.7
b_entity_trends/fixed_factor	ln_int	V3	171	-1.9	-3.9	+0.1
b_entity_trends/fixed_factor	ln_int	V12	30	-4.3	-8.9	+0.2
b_entity_trends/fixed_factor	ln_int	V17	13	-0.3	-3.1	+2.4
b_entity_trends/fixed_factor	ln_emis	V9	38	+0.4	-6.2	+6.9
b_entity_trends/fixed_factor	ln_emis	V4	122	-2.7	-11.6	+6.1
b_entity_trends/fixed_factor	ln_emis	V2	120	-0.0	-6.9	+6.9

Market concentration and verifier effects: supplementary material

Check	Outcome	Verifier	Switches	Effect	95\% low	95\% high
b_entity_trends/fixed_factor	ln_emis	V3	171	-3.8	-11.5	+3.9
b_entity_trends/fixed_factor	ln_emis	V12	30	+22.5	-1.4	+46.4
b_entity_trends/fixed_factor	ln_emis	V17	13	-21.3	-46.7	+4.1
b_entity_trends/fixed_factor	ln_energy	V9	38	+1.9	-4.6	+8.4
b_entity_trends/fixed_factor	ln_energy	V4	122	-1.6	-10.8	+7.5
b_entity_trends/fixed_factor	ln_energy	V2	120	+2.0	-4.6	+8.5
b_entity_trends/fixed_factor	ln_energy	V3	171	-1.9	-9.6	+5.8
b_entity_trends/fixed_factor	ln_energy	V12	30	+26.9	+3.8	+50.0
b_entity_trends/fixed_factor	ln_energy	V17	13	-21.0	-46.3	+4.4
d_quiet_movers/fixed_factor	ln_int	V9	26	+0.7	-1.1	+2.6
d_quiet_movers/fixed_factor	ln_int	V4	77	+1.8	-0.3	+3.9
d_quiet_movers/fixed_factor	ln_int	V2	76	+0.5	-1.2	+2.2
d_quiet_movers/fixed_factor	ln_int	V3	108	+0.7	-1.0	+2.4
d_quiet_movers/fixed_factor	ln_int	V12	12	+0.9	-5.4	+7.1
d_quiet_movers/fixed_factor	ln_int	V17	12	-0.5	-4.2	+3.1
d_quiet_movers/fixed_factor	ln_emis	V9	26	-3.4	-16.1	+9.4
d_quiet_movers/fixed_factor	ln_emis	V4	77	+12.7	-1.0	+26.4
d_quiet_movers/fixed_factor	ln_emis	V2	76	+7.3	-1.0	+15.7
d_quiet_movers/fixed_factor	ln_emis	V3	108	+6.7	-4.9	+18.3
d_quiet_movers/fixed_factor	ln_emis	V12	12	+8.4	-8.8	+25.6
d_quiet_movers/fixed_factor	ln_emis	V17	12	-6.3	-32.7	+20.2
d_quiet_movers/fixed_factor	ln_energy	V9	26	-4.1	-16.8	+8.7
d_quiet_movers/fixed_factor	ln_energy	V4	77	+10.9	-3.3	+25.1
d_quiet_movers/fixed_factor	ln_energy	V2	76	+6.8	-1.5	+15.1
d_quiet_movers/fixed_factor	ln_energy	V3	108	+6.0	-5.8	+17.9
d_quiet_movers/fixed_factor	ln_energy	V12	12	+7.5	-12.4	+27.5
d_quiet_movers/fixed_factor	ln_energy	V17	12	-5.7	-31.6	+20.1
e_all_entities/fixed_factor	ln_int	V9	40	+2.4	-1.2	+6.1
e_all_entities/fixed_factor	ln_int	V4	124	+3.5	+0.2	+6.8
e_all_entities/fixed_factor	ln_int	V2	123	+3.4	-0.0	+6.9
e_all_entities/fixed_factor	ln_int	V3	173	+2.3	-0.3	+4.9
e_all_entities/fixed_factor	ln_int	V12	32	-2.7	-6.8	+1.5
e_all_entities/fixed_factor	ln_int	V17	13	+1.5	-1.4	+4.4
e_all_entities/fixed_factor	ln_emis	V9	40	-10.3	-22.0	+1.4
e_all_entities/fixed_factor	ln_emis	V4	124	+1.4	-9.4	+12.2
e_all_entities/fixed_factor	ln_emis	V2	123	-0.8	-9.5	+7.8
e_all_entities/fixed_factor	ln_emis	V3	173	-3.5	-14.0	+7.0
e_all_entities/fixed_factor	ln_emis	V12	32	+6.6	-6.6	+19.9
e_all_entities/fixed_factor	ln_emis	V17	13	-21.1	-37.6	-4.5
e_all_entities/fixed_factor	ln_energy	V9	40	-12.7	-25.1	-0.4
e_all_entities/fixed_factor	ln_energy	V4	124	-2.1	-13.0	+8.9
e_all_entities/fixed_factor	ln_energy	V2	123	-4.3	-12.1	+3.5
e_all_entities/fixed_factor	ln_energy	V3	173	-5.8	-16.3	+4.7
e_all_entities/fixed_factor	ln_energy	V12	32	+9.3	-3.5	+22.0
e_all_entities/fixed_factor	ln_energy	V17	13	-22.5	-38.4	-6.7
f_years.2013_2019/fixed_factor	ln_int	V9	38	+1.5	-0.1	+3.2
f_years.2013_2019/fixed_factor	ln_int	V4	118	+2.1	-1.3	+5.6
f_years.2013_2019/fixed_factor	ln_int	V2	110	+0.3	-3.2	+3.8
f_years.2013_2019/fixed_factor	ln_int	V3	142	+1.9	-1.2	+4.9
f_years.2013_2019/fixed_factor	ln_int	V12	26	-5.1	-10.1	+0.0
f_years.2013_2019/fixed_factor	ln_emis	V9	38	-1.4	-6.8	+4.0
f_years.2013_2019/fixed_factor	ln_emis	V4	118	+9.2	+1.3	+17.0
f_years.2013_2019/fixed_factor	ln_emis	V2	110	+0.3	-6.2	+6.8
f_years.2013_2019/fixed_factor	ln_emis	V3	142	+0.4	-9.4	+10.2
f_years.2013_2019/fixed_factor	ln_emis	V12	26	+1.2	-8.7	+11.1
f_years.2013_2019/fixed_factor	ln_energy	V9	38	-2.9	-8.9	+3.1
f_years.2013_2019/fixed_factor	ln_energy	V4	118	+7.0	-1.7	+15.8
f_years.2013_2019/fixed_factor	ln_energy	V2	110	+0.0	-6.1	+6.2
f_years.2013_2019/fixed_factor	ln_energy	V3	142	-1.5	-11.3	+8.4
f_years.2013_2019/fixed_factor	ln_energy	V12	26	+6.2	-4.2	+16.7

Table S25: Change in ln energy at the switch by destination (in) and origin (out) verifier, fixed-factor movers

Direction	Verifier	Switches	Effect	s.e.	<i>t</i>
in	V13	11	-3.5	5.1	-0.69
in	V7	30	-1.7	8.8	-0.19
in	V5	25	-1.4	6.3	-0.22
in	V15	9	-0.8	5.2	-0.16
in	V8	9	-0.7	5.3	-0.14
in	V6	21	+0.6	14.2	0.04
in	V9	15	+2.2	6.8	0.32
in	V1	71	+2.4	5.1	0.47
in	V3	113	+3.9	4.9	0.80
in	V2	43	+5.2	5.7	0.90
in	V4	35	+7.3	6.0	1.21
in	V18	7	+10.1	16.9	0.60
in	V14	8	+14.0	6.8	2.08
in	V10	15	+25.2	13.5	1.87
in	V12	14	+36.0	17.5	2.06
out	V10	13	-13.2	14.2	-0.93
out	V6	11	-11.1	7.1	-1.57
out	V5	14	-7.4	5.7	-1.30
out	V12	15	-6.3	6.4	-0.98
out	V18	10	-5.1	6.3	-0.81
out	V8	9	-5.0	8.6	-0.58
out	V1	108	-4.7	4.3	-1.10
out	V17	10	-4.6	7.9	-0.58
out	V9	14	-3.2	5.5	-0.58
out	V4	69	-3.2	5.7	-0.56
out	V7	9	-1.0	6.7	-0.14
out	V14	6	-0.4	8.2	-0.05
out	V2	62	+0.8	6.0	0.14
out	V15	9	+1.2	4.1	0.30
out	V13	9	+1.5	7.3	0.20
out	V16	10	+1.7	5.8	0.29
out	V19	9	+1.8	7.0	0.25
out	V3	43	+4.5	7.4	0.60

Table S26: Industry-by-year cells

Sample	<i>n</i>	Cells	Singleton cells	Obs. in singletons	Share of obs. in cells of size 1–2	Median cell size
all movers	5449	428	62	62	0.025	8.000
fixed-factor	2441	236	39	39	0.031	6.000
discretion	2916	294	91	91	0.059	3.000

Table S27: Spearman rank correlations between published evaluation metrics and verifier effects: the 17 of 144 correlations with $p < 0.10$ (signs are mixed; no metric correlates consistently with the effects)

Round	Metric	Sample	Outcome	Effect measure	<i>n</i>	rho	<i>p</i>
2019-08	error_rate	fixed-factor	ln_emis	beta	14	-0.474	0.087
2019-08	error_rate	fixed-factor	ln_emis	abs_beta	14	-0.463	0.096
2019-08	error_rate	discretion	ln_int	abs_beta	15	-0.467	0.079
2019-08	error_rate	discretion	ln_energy	beta	15	-0.566	0.028
2019-08	correction_rate	all movers	ln_int	abs_beta	16	0.442	0.087

Round	Metric	Sample	Outcome	Effect measure	<i>n</i>	rho	<i>p</i>
2020-10	composite_score	all movers	ln_int	beta	15	0.513	0.051
2020-10	appropriateness_score	all movers	ln_emis	beta	15	0.448	0.094
2020-10	scale_score	fixed-factor	ln_int	beta	13	0.526	0.065
2020-10	scale_score	fixed-factor	ln_emis	abs_beta	13	-0.613	0.026
2020-10	scale_score	fixed-factor	ln_energy	abs_beta	13	-0.529	0.063
2020-10	scale_score	all movers	ln_int	beta	15	0.458	0.086
2020-10	operation_score	fixed-factor	ln_emis	beta	13	-0.528	0.064
2020-10	operation_score	fixed-factor	ln_energy	beta	13	-0.589	0.034
2020-10	operation_score	discretion	ln_int	beta	14	0.550	0.042
2020-10	operation_score	discretion	ln_int	abs_beta	14	-0.482	0.081
2020-10	operation_score	all movers	ln_int	beta	15	0.485	0.067
2020-10	operation_score	all movers	ln_int	abs_beta	15	-0.463	0.082

Exploratory: verifier effects on certified levels (95% CI)

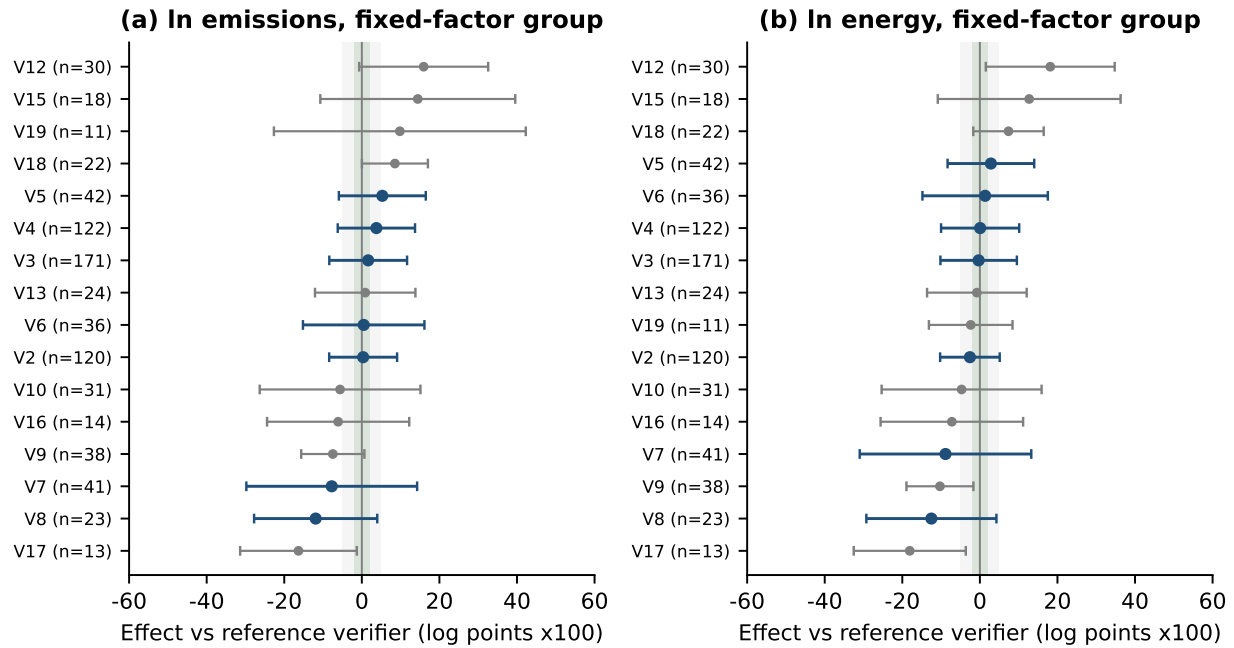


Figure S3: Exploratory: verifier effects on certified emission and energy levels, fixed-factor group (95% confidence intervals).

S7. Switch event study

The reading rule fixed in the analysis plan is applied: where the coefficient at $t - 1$ is at least as large as at t_0 , the text says that switching follows volatility. Cancellation-linked switches satisfy this for every outcome. Voluntary switches coincide with a higher probability of a large revision in certified energy at t_0 (5.4 percentage points, $p = 0.002$) with no elevation at $t - 1$; reverse causation (an energy-accounting change prompting the switch) is not excluded. Table S30 shows how the raw gap between switch years and other years in the probability of a large intensity revision (+6.3 percentage points) shrinks to +2.3 with entity and industry-by-year fixed effects and vanishes from 2015, outside waste and industrial-complex industries, and in the fixed-factor group; the pre-freeze feasibility figures are not used.

Table S28: Switch event study: coefficients at $t - 1$, t_0 and $t + 1$ for every sample, outcome and switch type (log points or percentage points $\times 100$; bootstrap p for t_0 in the scope-stable sample)

Sample	Outcome	Type	t	Events	Effect	s.e.	Boot. p
scope-stable	absd_ln_int	voluntary	-1	830	-0.59	0.65	
scope-stable	absd_ln_int	voluntary	0	830	+1.60	0.97	0.080
scope-stable	absd_ln_int	voluntary	1	830	+0.17	0.79	
scope-stable	absd_ln_int	exit_block	-1	80	+3.88	1.35	
scope-stable	absd_ln_int	exit_block	0	80	+3.95	2.42	0.124
scope-stable	absd_ln_int	exit_block	1	80	+7.59	7.27	
scope-stable	absd_ln_int	exit_dispersed	-1	28	+3.23	2.53	
scope-stable	absd_ln_int	exit_dispersed	0	28	+0.89	1.73	0.623
scope-stable	absd_ln_int	exit_dispersed	1	28	-0.62	1.62	
scope-stable	big_ln_int	voluntary	-1	830	-1.70	1.21	
scope-stable	big_ln_int	voluntary	0	830	+2.02	1.23	0.114
scope-stable	big_ln_int	voluntary	1	830	-0.46	1.27	
scope-stable	big_ln_int	exit_block	-1	80	+8.25	3.11	
scope-stable	big_ln_int	exit_block	0	80	+2.02	3.59	0.631
scope-stable	big_ln_int	exit_block	1	80	-2.82	2.58	
scope-stable	big_ln_int	exit_dispersed	-1	28	+5.32	7.76	
scope-stable	big_ln_int	exit_dispersed	0	28	+5.05	6.84	0.523
scope-stable	big_ln_int	exit_dispersed	1	28	-0.82	5.63	
scope-stable	absd_ln_emis	voluntary	-1	830	-1.87	0.98	
scope-stable	absd_ln_emis	voluntary	0	830	-0.12	1.11	0.924
scope-stable	absd_ln_emis	voluntary	1	830	+0.37	1.27	
scope-stable	absd_ln_emis	exit_block	-1	80	+1.38	2.39	
scope-stable	absd_ln_emis	exit_block	0	80	-0.92	2.62	0.727
scope-stable	absd_ln_emis	exit_block	1	80	+7.23	8.27	
scope-stable	absd_ln_emis	exit_dispersed	-1	28	+3.53	3.32	
scope-stable	absd_ln_emis	exit_dispersed	0	28	-0.42	3.39	0.898
scope-stable	absd_ln_emis	exit_dispersed	1	28	+4.29	3.88	
scope-stable	big_ln_emis	voluntary	-1	830	-0.80	1.93	
scope-stable	big_ln_emis	voluntary	0	830	+3.21	1.95	0.092
scope-stable	big_ln_emis	voluntary	1	830	-0.62	2.13	
scope-stable	big_ln_emis	exit_block	-1	80	+8.51	5.71	
scope-stable	big_ln_emis	exit_block	0	80	-0.18	5.71	0.968
scope-stable	big_ln_emis	exit_block	1	80	-2.40	5.74	
scope-stable	big_ln_emis	exit_dispersed	-1	28	+3.41	7.94	
scope-stable	big_ln_emis	exit_dispersed	0	28	+3.16	8.50	0.709
scope-stable	big_ln_emis	exit_dispersed	1	28	+19.13	7.35	
scope-stable	absd_ln_energy	voluntary	-1	830	-3.00	1.09	
scope-stable	absd_ln_energy	voluntary	0	830	+1.55	1.38	0.253
scope-stable	absd_ln_energy	voluntary	1	830	-1.33	1.32	
scope-stable	absd_ln_energy	exit_block	-1	80	+1.32	2.75	
scope-stable	absd_ln_energy	exit_block	0	80	-0.48	2.20	0.850
scope-stable	absd_ln_energy	exit_block	1	80	-0.22	2.24	
scope-stable	absd_ln_energy	exit_dispersed	-1	28	+1.71	3.68	
scope-stable	absd_ln_energy	exit_dispersed	0	28	-1.72	3.35	0.597
scope-stable	absd_ln_energy	exit_dispersed	1	28	+4.64	4.08	
scope-stable	big_ln_energy	voluntary	-1	830	-0.76	2.03	
scope-stable	big_ln_energy	voluntary	0	830	+5.35	1.95	0.002
scope-stable	big_ln_energy	voluntary	1	830	-0.71	2.00	
scope-stable	big_ln_energy	exit_block	-1	80	+10.78	5.85	
scope-stable	big_ln_energy	exit_block	0	80	+0.27	5.49	0.968
scope-stable	big_ln_energy	exit_block	1	80	-0.80	5.39	
scope-stable	big_ln_energy	exit_dispersed	-1	28	-0.94	8.92	
scope-stable	big_ln_energy	exit_dispersed	0	28	+2.07	9.01	0.838
scope-stable	big_ln_energy	exit_dispersed	1	28	+8.76	8.61	
all pairs	absd_ln_int	voluntary	-1	944	-0.15	0.64	
all pairs	absd_ln_int	voluntary	0	944	+0.93	0.76	
all pairs	absd_ln_int	voluntary	1	944	+0.07	0.66	
all pairs	absd_ln_int	exit_block	-1	110	+3.54	1.56	
all pairs	absd_ln_int	exit_block	0	110	+4.12	1.67	
all pairs	absd_ln_int	exit_block	1	110	+7.45	6.64	
all pairs	absd_ln_int	exit_dispersed	-1	36	+3.48	2.49	
all pairs	absd_ln_int	exit_dispersed	0	36	+1.32	1.89	

Sample	Outcome	Type	<i>t</i>	Events	Effect	s.e.	Boot. <i>p</i>
all pairs	absd_ln_int	exit_dispersed	1	36	+1.27	2.21	
all pairs	big_ln_int	voluntary	-1	944	-0.54	1.08	
all pairs	big_ln_int	voluntary	0	944	+1.86	1.22	
all pairs	big_ln_int	voluntary	1	944	-0.62	1.20	
all pairs	big_ln_int	exit_block	-1	110	+9.53	3.19	
all pairs	big_ln_int	exit_block	0	110	+4.20	3.10	
all pairs	big_ln_int	exit_block	1	110	-2.22	2.40	
all pairs	big_ln_int	exit_dispersed	-1	36	+2.86	7.22	
all pairs	big_ln_int	exit_dispersed	0	36	+5.08	6.78	
all pairs	big_ln_int	exit_dispersed	1	36	+0.01	5.54	
all pairs	absd_ln_emis	voluntary	-1	944	-2.34	0.82	
all pairs	absd_ln_emis	voluntary	0	944	-0.54	1.15	
all pairs	absd_ln_emis	voluntary	1	944	-0.70	1.67	
all pairs	absd_ln_emis	exit_block	-1	110	+2.59	2.18	
all pairs	absd_ln_emis	exit_block	0	110	-0.27	1.77	
all pairs	absd_ln_emis	exit_block	1	110	+7.15	7.85	
all pairs	absd_ln_emis	exit_dispersed	-1	36	+3.39	3.07	
all pairs	absd_ln_emis	exit_dispersed	0	36	+2.01	3.21	
all pairs	absd_ln_emis	exit_dispersed	1	36	+3.78	3.62	
all pairs	big_ln_emis	voluntary	-1	944	-0.97	1.81	
all pairs	big_ln_emis	voluntary	0	944	+2.79	1.88	
all pairs	big_ln_emis	voluntary	1	944	-0.19	2.05	
all pairs	big_ln_emis	exit_block	-1	110	+12.53	5.63	
all pairs	big_ln_emis	exit_block	0	110	-0.25	4.98	
all pairs	big_ln_emis	exit_block	1	110	-4.02	5.64	
all pairs	big_ln_emis	exit_dispersed	-1	36	+2.78	7.89	
all pairs	big_ln_emis	exit_dispersed	0	36	+9.91	8.69	
all pairs	big_ln_emis	exit_dispersed	1	36	+21.25	7.28	
all pairs	absd_ln_energy	voluntary	-1	944	-2.89	0.96	
all pairs	absd_ln_energy	voluntary	0	944	+0.59	1.33	
all pairs	absd_ln_energy	voluntary	1	944	-2.42	1.70	
all pairs	absd_ln_energy	exit_block	-1	110	+1.83	2.65	
all pairs	absd_ln_energy	exit_block	0	110	+0.24	1.98	
all pairs	absd_ln_energy	exit_block	1	110	-0.19	2.11	
all pairs	absd_ln_energy	exit_dispersed	-1	36	+2.57	3.35	
all pairs	absd_ln_energy	exit_dispersed	0	36	-1.10	2.96	
all pairs	absd_ln_energy	exit_dispersed	1	36	+6.02	4.31	
all pairs	big_ln_energy	voluntary	-1	944	-0.67	1.94	
all pairs	big_ln_energy	voluntary	0	944	+4.18	1.87	
all pairs	big_ln_energy	voluntary	1	944	-0.46	1.91	
all pairs	big_ln_energy	exit_block	-1	110	+11.82	5.55	
all pairs	big_ln_energy	exit_block	0	110	+1.37	4.62	
all pairs	big_ln_energy	exit_block	1	110	-2.45	5.45	
all pairs	big_ln_energy	exit_dispersed	-1	36	+2.31	8.31	
all pairs	big_ln_energy	exit_dispersed	0	36	+3.04	7.91	
all pairs	big_ln_energy	exit_dispersed	1	36	+11.77	8.28	
fixed-factor	absd_ln_int	voluntary	-1	396	+0.34	0.48	
fixed-factor	absd_ln_int	voluntary	0	396	-0.49	0.42	
fixed-factor	absd_ln_int	voluntary	1	396	+0.07	0.59	
fixed-factor	absd_ln_int	exit_block	-1	57	+2.02	1.52	
fixed-factor	absd_ln_int	exit_block	0	57	+1.47	1.72	
fixed-factor	absd_ln_int	exit_block	1	57	-0.75	0.98	
fixed-factor	absd_ln_int	exit_dispersed	-1	16	+4.44	3.12	
fixed-factor	absd_ln_int	exit_dispersed	0	16	+0.06	1.19	
fixed-factor	absd_ln_int	exit_dispersed	1	16	-0.24	1.18	
fixed-factor	big_ln_int	voluntary	-1	396	+0.79	1.19	
fixed-factor	big_ln_int	voluntary	0	396	+1.07	1.40	
fixed-factor	big_ln_int	voluntary	1	396	-0.72	1.08	
fixed-factor	big_ln_int	exit_block	-1	57	+7.14	3.70	
fixed-factor	big_ln_int	exit_block	0	57	+2.39	4.14	
fixed-factor	big_ln_int	exit_block	1	57	-1.89	2.07	
fixed-factor	big_ln_int	exit_dispersed	-1	16	+6.88	7.48	
fixed-factor	big_ln_int	exit_dispersed	0	16	+7.24	7.02	
fixed-factor	big_ln_int	exit_dispersed	1	16	+1.31	2.14	
fixed-factor	absd_ln_emis	voluntary	-1	396	-2.99	1.40	
fixed-factor	absd_ln_emis	voluntary	0	396	+0.40	1.71	

Sample	Outcome	Type	<i>t</i>	Events	Effect	s.e.	Boot. <i>p</i>
fixed-factor	absd_ln_emis	voluntary	1	396	-2.28	1.20	
fixed-factor	absd_ln_emis	exit_block	-1	57	+1.56	3.37	
fixed-factor	absd_ln_emis	exit_block	0	57	-3.81	1.83	
fixed-factor	absd_ln_emis	exit_block	1	57	-0.21	3.07	
fixed-factor	absd_ln_emis	exit_dispersed	-1	16	+3.86	5.00	
fixed-factor	absd_ln_emis	exit_dispersed	0	16	-2.45	3.20	
fixed-factor	absd_ln_emis	exit_dispersed	1	16	+6.48	5.25	
fixed-factor	big_ln_emis	voluntary	-1	396	-1.34	2.79	
fixed-factor	big_ln_emis	voluntary	0	396	-0.21	2.64	
fixed-factor	big_ln_emis	voluntary	1	396	-2.37	2.85	
fixed-factor	big_ln_emis	exit_block	-1	57	+6.45	7.40	
fixed-factor	big_ln_emis	exit_block	0	57	-12.51	6.44	
fixed-factor	big_ln_emis	exit_block	1	57	-8.66	7.73	
fixed-factor	big_ln_emis	exit_dispersed	-1	16	+8.96	9.10	
fixed-factor	big_ln_emis	exit_dispersed	0	16	-2.21	13.84	
fixed-factor	big_ln_emis	exit_dispersed	1	16	+22.97	10.60	
fixed-factor	absd_ln_energy	voluntary	-1	396	-3.29	1.45	
fixed-factor	absd_ln_energy	voluntary	0	396	+0.33	1.75	
fixed-factor	absd_ln_energy	voluntary	1	396	-2.54	1.26	
fixed-factor	absd_ln_energy	exit_block	-1	57	+2.37	3.61	
fixed-factor	absd_ln_energy	exit_block	0	57	-1.26	2.46	
fixed-factor	absd_ln_energy	exit_block	1	57	+0.09	3.11	
fixed-factor	absd_ln_energy	exit_dispersed	-1	16	+1.17	4.02	
fixed-factor	absd_ln_energy	exit_dispersed	0	16	-2.15	2.86	
fixed-factor	absd_ln_energy	exit_dispersed	1	16	+6.27	5.21	
fixed-factor	big_ln_energy	voluntary	-1	396	-4.31	2.80	
fixed-factor	big_ln_energy	voluntary	0	396	+0.48	2.58	
fixed-factor	big_ln_energy	voluntary	1	396	-1.94	2.90	
fixed-factor	big_ln_energy	exit_block	-1	57	+6.12	7.59	
fixed-factor	big_ln_energy	exit_block	0	57	-7.83	6.33	
fixed-factor	big_ln_energy	exit_block	1	57	-4.79	8.04	
fixed-factor	big_ln_energy	exit_dispersed	-1	16	+3.98	8.15	
fixed-factor	big_ln_energy	exit_dispersed	0	16	+1.24	15.31	
fixed-factor	big_ln_energy	exit_dispersed	1	16	+23.32	12.55	
discretion	absd_ln_int	voluntary	-1	548	-0.58	1.09	
discretion	absd_ln_int	voluntary	0	548	+2.18	1.33	
discretion	absd_ln_int	voluntary	1	548	+0.32	1.09	
discretion	absd_ln_int	exit_block	-1	53	+2.96	3.35	
discretion	absd_ln_int	exit_block	0	53	+6.24	2.86	
discretion	absd_ln_int	exit_block	1	53	+15.68	13.49	
discretion	absd_ln_int	exit_dispersed	-1	20	+2.98	3.15	
discretion	absd_ln_int	exit_dispersed	0	20	+2.56	3.05	
discretion	absd_ln_int	exit_dispersed	1	20	+2.72	4.04	
discretion	big_ln_int	voluntary	-1	548	-1.68	1.74	
discretion	big_ln_int	voluntary	0	548	+2.66	1.93	
discretion	big_ln_int	voluntary	1	548	-0.50	1.98	
discretion	big_ln_int	exit_block	-1	53	+10.79	5.30	
discretion	big_ln_int	exit_block	0	53	+4.82	4.70	
discretion	big_ln_int	exit_block	1	53	-3.70	4.50	
discretion	big_ln_int	exit_dispersed	-1	20	+0.33	11.28	
discretion	big_ln_int	exit_dispersed	0	20	+2.77	10.27	
discretion	big_ln_int	exit_dispersed	1	20	+1.22	9.88	
discretion	absd_ln_emis	voluntary	-1	548	-2.31	1.23	
discretion	absd_ln_emis	voluntary	0	548	-1.35	1.35	
discretion	absd_ln_emis	voluntary	1	548	+0.84	2.87	
discretion	absd_ln_emis	exit_block	-1	53	+3.48	2.91	
discretion	absd_ln_emis	exit_block	0	53	+3.46	3.26	
discretion	absd_ln_emis	exit_block	1	53	+15.12	15.73	
discretion	absd_ln_emis	exit_dispersed	-1	20	-0.63	4.44	
discretion	absd_ln_emis	exit_dispersed	0	20	+4.78	5.22	
discretion	absd_ln_emis	exit_dispersed	1	20	-2.36	5.45	
discretion	big_ln_emis	voluntary	-1	548	-1.04	2.48	
discretion	big_ln_emis	voluntary	0	548	+5.67	2.68	
discretion	big_ln_emis	voluntary	1	548	+2.61	2.98	
discretion	big_ln_emis	exit_block	-1	53	+18.45	8.45	
discretion	big_ln_emis	exit_block	0	53	+10.72	7.10	

Sample	Outcome	Type	t	Events	Effect	s.e.	Boot. p
discretion	big_ln_emis	exit_block	1	53	-0.79	8.29	
discretion	big_ln_emis	exit_dispersed	-1	20	-6.03	12.03	
discretion	big_ln_emis	exit_dispersed	0	20	+19.73	11.37	
discretion	big_ln_emis	exit_dispersed	1	20	+20.12	11.06	
discretion	absd_ln_energy	voluntary	-1	548	-2.50	1.43	
discretion	absd_ln_energy	voluntary	0	548	+0.76	1.75	
discretion	absd_ln_energy	voluntary	1	548	-2.22	2.96	
discretion	absd_ln_energy	exit_block	-1	53	-0.32	4.53	
discretion	absd_ln_energy	exit_block	0	53	+1.76	3.38	
discretion	absd_ln_energy	exit_block	1	53	-0.20	3.24	
discretion	absd_ln_energy	exit_dispersed	-1	20	+1.84	5.49	
discretion	absd_ln_energy	exit_dispersed	0	20	+0.22	4.84	
discretion	absd_ln_energy	exit_dispersed	1	20	+3.58	7.60	
discretion	big_ln_energy	voluntary	-1	548	+1.75	2.74	
discretion	big_ln_energy	voluntary	0	548	+6.92	2.74	
discretion	big_ln_energy	voluntary	1	548	+1.11	2.60	
discretion	big_ln_energy	exit_block	-1	53	+17.67	8.00	
discretion	big_ln_energy	exit_block	0	53	+10.04	6.28	
discretion	big_ln_energy	exit_block	1	53	-0.26	7.40	
discretion	big_ln_energy	exit_dispersed	-1	20	-1.67	13.39	
discretion	big_ln_energy	exit_dispersed	0	20	+5.69	8.48	
discretion	big_ln_energy	exit_dispersed	1	20	+1.65	11.55	
years $\geq$ 2015	absd_ln_int	voluntary	-1	768	-0.12	0.70	
years $\geq$ 2015	absd_ln_int	voluntary	0	768	+1.14	0.81	
years $\geq$ 2015	absd_ln_int	voluntary	1	768	-0.19	0.76	
years $\geq$ 2015	absd_ln_int	exit_block	-1	110	+3.01	1.64	
years $\geq$ 2015	absd_ln_int	exit_block	0	110	+3.62	1.72	
years $\geq$ 2015	absd_ln_int	exit_block	1	110	+6.95	6.52	
years $\geq$ 2015	absd_ln_int	exit_dispersed	-1	36	+3.33	2.44	
years $\geq$ 2015	absd_ln_int	exit_dispersed	0	36	+1.36	2.15	
years $\geq$ 2015	absd_ln_int	exit_dispersed	1	36	+1.21	2.37	
years $\geq$ 2015	big_ln_int	voluntary	-1	768	-0.86	1.15	
years $\geq$ 2015	big_ln_int	voluntary	0	768	+0.34	1.36	
years $\geq$ 2015	big_ln_int	voluntary	1	768	-1.75	1.20	
years $\geq$ 2015	big_ln_int	exit_block	-1	110	+8.05	3.08	
years $\geq$ 2015	big_ln_int	exit_block	0	110	+2.67	2.95	
years $\geq$ 2015	big_ln_int	exit_block	1	110	-3.76	2.26	
years $\geq$ 2015	big_ln_int	exit_dispersed	-1	36	+3.04	7.46	
years $\geq$ 2015	big_ln_int	exit_dispersed	0	36	+5.36	7.32	
years $\geq$ 2015	big_ln_int	exit_dispersed	1	36	+0.46	6.14	
years $\geq$ 2015	absd_ln_emis	voluntary	-1	768	-2.42	0.91	
years $\geq$ 2015	absd_ln_emis	voluntary	0	768	-0.38	1.22	
years $\geq$ 2015	absd_ln_emis	voluntary	1	768	-0.79	2.11	
years $\geq$ 2015	absd_ln_emis	exit_block	-1	110	+2.60	2.24	
years $\geq$ 2015	absd_ln_emis	exit_block	0	110	-0.21	1.93	
years $\geq$ 2015	absd_ln_emis	exit_block	1	110	+7.20	8.02	
years $\geq$ 2015	absd_ln_emis	exit_dispersed	-1	36	+2.27	2.97	
years $\geq$ 2015	absd_ln_emis	exit_dispersed	0	36	+1.31	3.20	
years $\geq$ 2015	absd_ln_emis	exit_dispersed	1	36	+2.98	3.56	
years $\geq$ 2015	big_ln_emis	voluntary	-1	768	-1.78	1.91	
years $\geq$ 2015	big_ln_emis	voluntary	0	768	+1.34	2.08	
years $\geq$ 2015	big_ln_emis	voluntary	1	768	+0.41	2.24	
years $\geq$ 2015	big_ln_emis	exit_block	-1	110	+13.21	5.63	
years $\geq$ 2015	big_ln_emis	exit_block	0	110	+0.52	4.99	
years $\geq$ 2015	big_ln_emis	exit_block	1	110	-3.37	5.73	
years $\geq$ 2015	big_ln_emis	exit_dispersed	-1	36	+2.53	8.44	
years $\geq$ 2015	big_ln_emis	exit_dispersed	0	36	+10.12	9.13	
years $\geq$ 2015	big_ln_emis	exit_dispersed	1	36	+21.49	7.47	
years $\geq$ 2015	absd_ln_energy	voluntary	-1	768	-3.12	1.08	
years $\geq$ 2015	absd_ln_energy	voluntary	0	768	+0.76	1.44	
years $\geq$ 2015	absd_ln_energy	voluntary	1	768	-2.75	2.18	
years $\geq$ 2015	absd_ln_energy	exit_block	-1	110	+2.09	2.86	
years $\geq$ 2015	absd_ln_energy	exit_block	0	110	+0.58	2.18	
years $\geq$ 2015	absd_ln_energy	exit_block	1	110	+0.16	2.38	
years $\geq$ 2015	absd_ln_energy	exit_dispersed	-1	36	+1.04	3.36	
years $\geq$ 2015	absd_ln_energy	exit_dispersed	0	36	-2.40	2.73	

Sample	Outcome	Type	t	Events	Effect	s.e.	Boot. p
years $\geq$ 2015	absd_ln_energy	exit_dispersed	1	36	+4.60	4.12	
years $\geq$ 2015	big_ln_energy	voluntary	-1	768	-1.71	2.01	
years $\geq$ 2015	big_ln_energy	voluntary	0	768	+2.56	2.03	
years $\geq$ 2015	big_ln_energy	voluntary	1	768	+0.14	2.13	
years $\geq$ 2015	big_ln_energy	exit_block	-1	110	+14.57	5.75	
years $\geq$ 2015	big_ln_energy	exit_block	0	110	+4.26	4.75	
years $\geq$ 2015	big_ln_energy	exit_block	1	110	+0.27	5.73	
years $\geq$ 2015	big_ln_energy	exit_dispersed	-1	36	-3.28	8.92	
years $\geq$ 2015	big_ln_energy	exit_dispersed	0	36	-3.01	7.83	
years $\geq$ 2015	big_ln_energy	exit_dispersed	1	36	+5.94	8.44	
excl. waste / industrial complex	absd_ln_int	voluntary	-1	789	+0.05	0.47	
excl. waste / industrial complex	absd_ln_int	voluntary	0	789	-0.89	0.49	
excl. waste / industrial complex	absd_ln_int	voluntary	1	789	-0.48	0.58	
excl. waste / industrial complex	absd_ln_int	exit_block	-1	97	+1.79	1.18	
excl. waste / industrial complex	absd_ln_int	exit_block	0	97	+1.38	1.21	
excl. waste / industrial complex	absd_ln_int	exit_block	1	97	-0.09	0.76	
excl. waste / industrial complex	absd_ln_int	exit_dispersed	-1	29	+2.11	2.16	
excl. waste / industrial complex	absd_ln_int	exit_dispersed	0	29	-0.71	1.15	
excl. waste / industrial complex	absd_ln_int	exit_dispersed	1	29	-0.18	1.47	
excl. waste / industrial complex	big_ln_int	voluntary	-1	789	-0.29	0.89	
excl. waste / industrial complex	big_ln_int	voluntary	0	789	+0.27	1.08	
excl. waste / industrial complex	big_ln_int	voluntary	1	789	-1.56	1.07	
excl. waste / industrial complex	big_ln_int	exit_block	-1	97	+6.17	2.89	
excl. waste / industrial complex	big_ln_int	exit_block	0	97	+1.80	2.81	
excl. waste / industrial complex	big_ln_int	exit_block	1	97	-2.10	1.58	
excl. waste / industrial complex	big_ln_int	exit_dispersed	-1	29	+0.29	5.03	
excl. waste / industrial complex	big_ln_int	exit_dispersed	0	29	+4.97	6.29	
excl. waste / industrial complex	big_ln_int	exit_dispersed	1	29	-0.48	4.09	
excl. waste / industrial complex	absd_ln_emis	voluntary	-1	789	-2.74	0.88	
excl. waste / industrial complex	absd_ln_emis	voluntary	0	789	-1.51	1.29	
excl. waste / industrial complex	absd_ln_emis	voluntary	1	789	-2.23	1.88	
excl. waste / industrial complex	absd_ln_emis	exit_block	-1	97	+1.36	2.44	
excl. waste / industrial complex	absd_ln_emis	exit_block	0	97	-2.44	1.49	
excl. waste / industrial complex	absd_ln_emis	exit_block	1	97	-0.36	2.10	
excl. waste / industrial complex	absd_ln_emis	exit_dispersed	-1	29	+1.75	3.44	
excl. waste / industrial complex	absd_ln_emis	exit_dispersed	0	29	-1.23	3.54	
excl. waste / industrial complex	absd_ln_emis	exit_dispersed	1	29	+4.43	4.09	
excl. waste / industrial complex	big_ln_emis	voluntary	-1	789	-0.35	1.99	
excl. waste / industrial complex	big_ln_emis	voluntary	0	789	+2.74	2.07	
excl. waste / industrial complex	big_ln_emis	voluntary	1	789	-0.29	2.27	
excl. waste / industrial complex	big_ln_emis	exit_block	-1	97	+12.01	6.06	
excl. waste / industrial complex	big_ln_emis	exit_block	0	97	-4.42	4.98	
excl. waste / industrial complex	big_ln_emis	exit_block	1	97	-4.62	5.96	
excl. waste / industrial complex	big_ln_emis	exit_dispersed	-1	29	+3.68	8.67	
excl. waste / industrial complex	big_ln_emis	exit_dispersed	0	29	+10.04	9.60	
excl. waste / industrial complex	big_ln_emis	exit_dispersed	1	29	+18.91	8.08	
excl. waste / industrial complex	absd_ln_energy	voluntary	-1	789	-3.32	1.01	
excl. waste / industrial complex	absd_ln_energy	voluntary	0	789	-1.63	1.39	
excl. waste / industrial complex	absd_ln_energy	voluntary	1	789	-3.23	1.98	
excl. waste / industrial complex	absd_ln_energy	exit_block	-1	97	+1.72	2.63	
excl. waste / industrial complex	absd_ln_energy	exit_block	0	97	-0.71	1.81	
excl. waste / industrial complex	absd_ln_energy	exit_block	1	97	+0.29	2.10	
excl. waste / industrial complex	absd_ln_energy	exit_dispersed	-1	29	-0.63	3.14	
excl. waste / industrial complex	absd_ln_energy	exit_dispersed	0	29	-3.07	3.61	
excl. waste / industrial complex	absd_ln_energy	exit_dispersed	1	29	+5.48	4.72	
excl. waste / industrial complex	big_ln_energy	voluntary	-1	789	-1.48	2.02	
excl. waste / industrial complex	big_ln_energy	voluntary	0	789	+2.95	1.98	
excl. waste / industrial complex	big_ln_energy	voluntary	1	789	-0.96	2.08	
excl. waste / industrial complex	big_ln_energy	exit_block	-1	97	+10.18	5.83	
excl. waste / industrial complex	big_ln_energy	exit_block	0	97	-0.90	4.79	
excl. waste / industrial complex	big_ln_energy	exit_block	1	97	-3.99	5.80	
excl. waste / industrial complex	big_ln_energy	exit_dispersed	-1	29	-1.20	8.80	
excl. waste / industrial complex	big_ln_energy	exit_dispersed	0	29	+6.36	9.77	
excl. waste / industrial complex	big_ln_energy	exit_dispersed	1	29	+18.89	9.00	

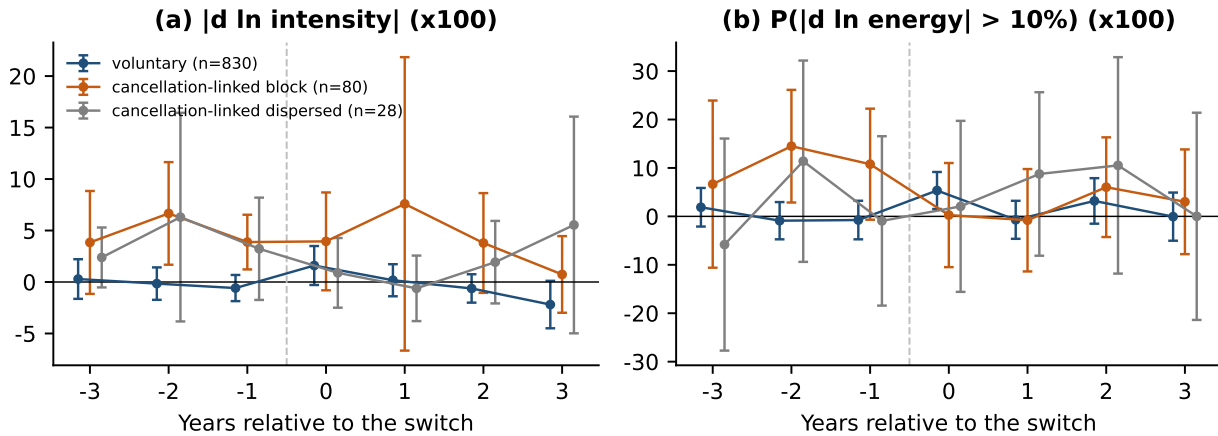
Table S29: Realised versus design MDE for the t_0 coefficients, scope-stable pairs

Outcome	Type	Events	Effect	s.e.	Boot. p	Realised MDE	Design MDE
absd_ln.int	voluntary	830	+1.60	0.97	0.080	2.70	2.24
absd_ln.int	exit_block	80	+3.95	2.42	0.124	6.79	4.75
absd_ln.int	exit_dispersed	28	+0.89	1.73	0.623	4.83	3.94
big_ln.int	voluntary	830	+2.02	1.23	0.114	3.45	2.24
big_ln.int	exit_block	80	+2.02	3.59	0.631	10.06	4.75
big_ln.int	exit_dispersed	28	+5.05	6.84	0.523	19.16	3.94
absd_ln.emis	voluntary	830	-0.12	1.11	0.924	3.09	2.24
absd_ln.emis	exit_block	80	-0.92	2.62	0.727	7.32	4.75
absd_ln.emis	exit_dispersed	28	-0.42	3.39	0.898	9.49	3.94
big_ln.emis	voluntary	830	+3.21	1.95	0.092	5.47	2.24
big_ln.emis	exit_block	80	-0.18	5.71	0.968	15.99	4.75
big_ln.emis	exit_dispersed	28	+3.16	8.50	0.709	23.80	3.94
absd_ln.energy	voluntary	830	+1.55	1.38	0.253	3.86	2.24
absd_ln.energy	exit_block	80	-0.48	2.20	0.850	6.15	4.75
absd_ln.energy	exit_dispersed	28	-1.72	3.35	0.597	9.38	3.94
big_ln.energy	voluntary	830	+5.35	1.95	0.002	5.46	2.24
big_ln.energy	exit_block	80	+0.27	5.49	0.968	15.36	4.75
big_ln.energy	exit_dispersed	28	+2.07	9.01	0.838	25.22	3.94

Table S30: From the raw to the conditional gap in the probability of a large intensity revision in switch years

Step	Gap (pp)	s.e.	Boot. p	Switch years	n	Note
1 raw means (all pairs 2013+)	+6.27			1090	10368	switch years 0.167 vs other years 0.104
2 + entity FE + industry x year FE	+2.31	1.06	0.026	1090	10368	
3 years $\geq$ 2015	+1.18	1.15	0.285	914	9268	
4 excluding waste / industrial complexes	+0.65	0.93	0.477	915	9355	
5 scope-stable pairs	+2.30	1.07	0.028	938	9199	
6 fixed-factor group	+0.76	1.13	0.485	469	4834	
7 switch at t-1 (lag), all pairs	+0.67	0.97	0.523	977	10368	

Switch event study, scope-stable pairs (entity FE + industry x year FE, 95% CI)

**Figure S4:** Switch event study on scope-stable pairs, $t-3$ to $t+3$, by switch type (95% confidence intervals).

S8. Block-transfer difference-in-differences

Table S31: Block-transfer difference-in-differences: all event-time coefficients relative to the year before the transfer (log points $\times 100$); pre-trend tests, post-period averages and equivalence flags are in Table 4 of the main text

Event (treated n)	Control (n)	Outcome	Event time	Effect	s.e.
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_int	T-3	+9.5	13.4
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_int	T-2	+8.7	10.0
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_int	T+0	+1.8	3.2
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_int	T+1	+1.0	4.4
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_int	T+2	-1.4	4.9
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_emis	T-3	+9.6	11.9
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_emis	T-2	+9.7	11.2
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_emis	T+0	-0.7	4.3
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_emis	T+1	-5.2	3.5
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_emis	T+2	-10.5	8.7
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_energy	T-3	+0.1	7.3
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_energy	T-2	+1.0	4.8
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_energy	T+0	-2.5	4.7
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_energy	T+1	-6.2	4.9
V4->V3 2022 ($n=71$)	successor_stayers ($n=121$)	ln_energy	T+2	-9.2	10.3
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_int	T-3	+14.0	15.1
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_int	T-2	+10.9	11.3
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_int	T+0	+2.8	3.3
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_int	T+1	-2.1	3.4
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_int	T+2	+1.1	3.3
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_emis	T-3	+9.3	11.9
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_emis	T-2	+9.7	12.1
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_emis	T+0	-0.6	1.8
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_emis	T+1	-2.9	3.7
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_emis	T+2	-14.1	10.2
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_energy	T-3	-4.6	5.2
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_energy	T-2	-1.3	3.3
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_energy	T+0	-3.3	3.1
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_energy	T+1	-0.9	4.4
V4->V3 2022 ($n=71$)	all_stayers ($n=794$)	ln_energy	T+2	-15.2	11.4
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T-3	+0.2	0.4
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T-2	+0.2	0.4
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T+0	+0.1	0.4
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T+1	-0.3	0.5
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T+2	-1.0	0.7
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_int	T+3	-1.6	0.8
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T-3	-19.8	7.1
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T-2	-17.0	6.8
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T+0	+6.1	7.8
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T+1	-2.6	4.2
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T+2	+0.3	4.9
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_emis	T+3	+0.3	5.2
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T-3	-20.0	7.0
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T-2	-17.2	6.7
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T+0	+6.0	7.6
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T+1	-2.3	4.1
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T+2	+1.3	4.9
V17->V3 2021 ($n=14$)	successor_stayers ($n=121$)	ln_energy	T+3	+1.9	5.3
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T-3	-0.2	0.5
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T-2	+0.0	0.4
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T+0	+0.0	0.5
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T+1	+0.9	0.5
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T+2	+0.1	0.7
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_int	T+3	+0.5	0.7
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T-3	-14.6	5.0
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T-2	-10.0	4.9
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T+0	+5.2	4.9
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T+1	+2.4	4.6

Event (treated n)	Control (n)	Outcome	Event time	Effect	s.e.
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T+2	+8.7	5.5
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_emis	T+3	+11.3	6.4
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T-3	-14.4	4.8
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T-2	-10.0	4.8
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T+0	+5.2	4.8
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T+1	+1.5	4.5
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T+2	+8.7	5.4
V17->V3 2021 ($n=14$)	all_stayers ($n=930$)	ln_energy	T+3	+10.8	6.4
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T-3	+0.6	1.1
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T-2	-0.5	1.2
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T+0	-0.5	1.4
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T+1	-2.6	3.3
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T+2	+1.0	0.9
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_int	T+3	+4.6	3.3
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T-3	+11.6	9.7
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T-2	-0.4	2.8
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T+0	-0.2	2.7
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T+1	+0.6	5.5
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T+2	+8.3	8.5
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_emis	T+3	+12.3	7.3
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T-3	+11.0	9.5
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T-2	+0.1	2.9
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T+0	+0.3	1.9
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T+1	+3.2	6.7
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T+2	+7.3	8.6
V18->V1 2018 ($n=13$)	successor_stayers ($n=273$)	ln_energy	T+3	+7.7	6.3
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T-3	+1.1	1.1
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T-2	+0.1	1.4
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T+0	+1.1	2.4
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T+1	-2.1	3.2
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T+2	+2.2	1.6
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_int	T+3	+5.3	3.3
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T-3	+4.3	5.7
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T-2	+4.5	6.5
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T+0	-2.8	3.0
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T+1	-7.0	4.3
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T+2	-4.8	5.5
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_emis	T+3	+0.2	7.5
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T-3	+3.2	5.5
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T-2	+4.4	6.6
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T+0	-3.9	2.8
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T+1	-4.9	5.4
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T+2	-6.9	5.5
V18->V1 2018 ($n=13$)	all_stayers ($n=797$)	ln_energy	T+3	-5.1	6.2

S9. Evaluation disclosures and client reallocation

The frozen hypothesis is that the verifiers in the bottom tercile of each published metric lose more clients (net outflow in the next verification season) than those in the top tercile, with first effect seasons 2020 (2019 disclosure) and 2021 (2020 disclosure). The 2019 disclosure covered the 2018 statements of ETS entities (per-verifier verification counts match the registry's 2018 ETS client counts with correlation 1.000), so the verifier-season panel is built from ETS clients. The frozen script broke ties in the metric by sort order; the 2019 error rate has a tie at the fifth-worst position and the 2020 score a tie at the fifth-best position, so Table S34 re-estimates every tie resolution and Table S35 uses the published rank with all tied verifiers included. Randomisation inference enumerates every assignment of the treated set among the verifiers in the sample. The entity-level linear-probability model with leads that the plan listed was not run; the verifier-season

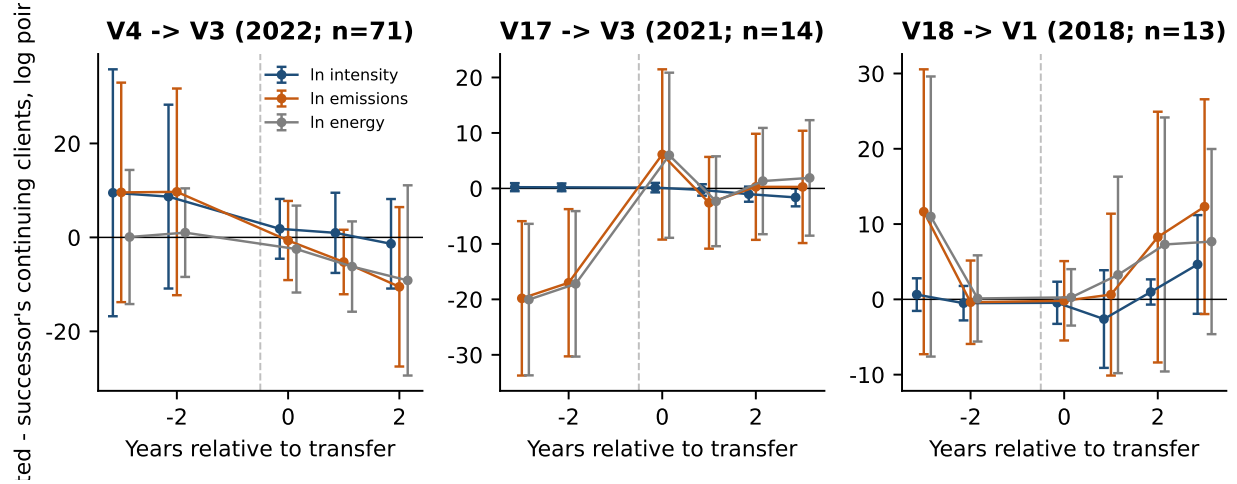


Figure S5: Block transfers: treated minus the successor's continuing clients, by event time (95% confidence intervals).

randomisation inference and leave-one-out checks replace it (logged in S11).

Table S32: Published evaluation metrics used for the terciles (2019: error rate, lower is better; 2020: composite score, higher is better)

Round	Metric	Verifier	Value	Published rank
2019-08	error_rate	V12	1.0	1
2019-08	error_rate	V5	2.0	2
2019-08	error_rate	V14	2.0	3
2019-08	error_rate	V15	3.0	4
2019-08	error_rate	V7	5.0	5
2019-08	error_rate	V3	7.0	6
2019-08	error_rate	V4	7.0	7
2019-08	error_rate	V16	8.0	8
2019-08	error_rate	V1	8.0	9
2019-08	error_rate	V8	9.0	10
2019-08	error_rate	V17	11.0	11
2019-08	error_rate	V2	12.0	12
2019-08	error_rate	V9	12.0	13
2019-08	error_rate	V6	13.0	14
2019-08	error_rate	V10	17.0	15
2019-08	error_rate	V13	22.0	16
2019-08	error_rate	V11	23.0	17
2020-10	composite_score	V9	92.2	1
2020-10	composite_score	V13	91.6	2
2020-10	composite_score	V8	90.6	3
2020-10	composite_score	V4	90.0	4
2020-10	composite_score	V6	88.6	5
2020-10	composite_score	V14	88.6	5
2020-10	composite_score	V11	88.4	7
2020-10	composite_score	V3	87.8	8
2020-10	composite_score	V17	86.6	9
2020-10	composite_score	V12	86.4	10
2020-10	composite_score	V5	85.8	11
2020-10	composite_score	V1	85.4	12
2020-10	composite_score	V2	83.6	13

Round	Metric	Verifier	Value	Published rank
2020-10	composite_score	V7	69.8	14
2020-10	composite_score	V15	65.0	15
2020-10	composite_score	V10	48.6	16

Table S33: Frozen test and pre-period placebo (verifier-season DiD, ETS clients)

Test	Placebo	Outcome	Effect	s.e.	Boot. <i>p</i>	<i>G</i>	<i>n</i>	Treated
2019-08 error_rate bottom vs top tercile	0	ln_clients	+0.089	0.213	0.674	10	69	5
2019-08 error_rate bottom vs top tercile	0	outflow_rate	+0.054	0.054	0.329	10	69	5
2019-08 error_rate bottom vs top tercile	0	inflow_rate	-0.076	0.086	0.422	10	69	5
2019-08 error_rate bottom vs top tercile	0	net_rate	-0.130	0.122	0.303	10	69	5
2019-08 error_rate bottom vs top tercile	1	ln_clients	-0.088	0.164	0.601	10	30	5
2019-08 error_rate bottom vs top tercile	1	outflow_rate	+0.019	0.121	0.870	10	30	5
2019-08 error_rate bottom vs top tercile	1	inflow_rate	-0.044	0.069	0.560	10	30	5
2019-08 error_rate bottom vs top tercile	1	net_rate	-0.063	0.164	0.707	10	30	5
2020-10 composite_score bottom vs top tercile	0	ln_clients	-0.283	0.119	0.049	10	68	5
2020-10 composite_score bottom vs top tercile	0	outflow_rate	-0.036	0.031	0.283	10	68	5
2020-10 composite_score bottom vs top tercile	0	inflow_rate	-0.077	0.033	0.041	10	68	5
2020-10 composite_score bottom vs top tercile	0	net_rate	-0.041	0.040	0.320	10	68	5
2020-10 composite_score bottom vs top tercile	1	ln_clients	-0.011	0.146	0.957	10	40	5
2020-10 composite_score bottom vs top tercile	1	outflow_rate	+0.150	0.074	0.079	10	40	5
2020-10 composite_score bottom vs top tercile	1	inflow_rate	+0.038	0.073	0.627	10	40	5
2020-10 composite_score bottom vs top tercile	1	net_rate	-0.112	0.127	0.461	10	40	5

Table S34: Tercile tie sensitivity

Round	Configuration	Outcome	Effect	s.e.	Boot. <i>p</i>	<i>G</i>	<i>n</i>
2019-08	value_sort_as_frozen	ln_clients	+0.089	0.213	0.674	10	69
2019-08	value_sort_as_frozen	outflow_rate	+0.054	0.054	0.329	10	69
2019-08	value_sort_as_frozen	inflow_rate	-0.076	0.086	0.422	10	69
2019-08	value_sort_as_frozen	net_rate	-0.130	0.122	0.303	10	69
2019-08	tie_DAEIL_in_bottom	ln_clients	+0.075	0.213	0.722	10	69
2019-08	tie_DAEIL_in_bottom	outflow_rate	+0.043	0.052	0.448	10	69
2019-08	tie_DAEIL_in_bottom	inflow_rate	-0.057	0.086	0.548	10	69
2019-08	tie_DAEIL_in_bottom	net_rate	-0.100	0.118	0.435	10	69
2019-08	published_rank	ln_clients	+0.075	0.213	0.722	10	69
2019-08	published_rank	outflow_rate	+0.043	0.052	0.448	10	69
2019-08	published_rank	inflow_rate	-0.057	0.086	0.548	10	69
2019-08	published_rank	net_rate	-0.100	0.118	0.435	10	69
2019-08	drop_tied_pair	ln_clients	+0.101	0.231	0.664	9	62
2019-08	drop_tied_pair	outflow_rate	+0.038	0.056	0.529	9	62
2019-08	drop_tied_pair	inflow_rate	-0.065	0.093	0.527	9	62
2019-08	drop_tied_pair	net_rate	-0.103	0.128	0.469	9	62
2020-10	value_sort_as_frozen	ln_clients	-0.283	0.119	0.049	10	68
2020-10	value_sort_as_frozen	outflow_rate	-0.036	0.031	0.283	10	68
2020-10	value_sort_as_frozen	inflow_rate	-0.077	0.033	0.041	10	68
2020-10	value_sort_as_frozen	net_rate	-0.041	0.040	0.320	10	68
2020-10	tie_SUSTAIN_in_top	ln_clients	-0.338	0.143	0.028	10	68
2020-10	tie_SUSTAIN_in_top	outflow_rate	-0.046	0.026	0.086	10	68
2020-10	tie_SUSTAIN_in_top	inflow_rate	-0.174	0.096	0.057	10	68
2020-10	tie_SUSTAIN_in_top	net_rate	-0.128	0.107	0.283	10	68
2020-10	both_tied_in_top	ln_clients	-0.359	0.131	0.016	11	75
2020-10	both_tied_in_top	outflow_rate	-0.030	0.029	0.312	11	75
2020-10	both_tied_in_top	inflow_rate	-0.157	0.081	0.050	11	75
2020-10	both_tied_in_top	net_rate	-0.127	0.090	0.182	11	75
2020-10	drop_tied_pair	ln_clients	-0.236	0.118	0.098	9	61
2020-10	drop_tied_pair	outflow_rate	-0.058	0.025	0.040	9	61
2020-10	drop_tied_pair	inflow_rate	-0.077	0.037	0.080	9	61
2020-10	drop_tied_pair	net_rate	-0.019	0.037	0.630	9	61

Table S35: Robustness set: published-rank terciles, wider control group, randomisation inference (RI), common placebo window, controls and realised MDE

Round	Check	Configuration	Outcome	Effect	s.e.	Boot. p	RI p	MDE	G
2019-08	realised mde	frozen	ln_clients	+0.089	0.213			0.596	10
2019-08	realised mde	frozen	outflow_rate	+0.054	0.054			0.150	10
2019-08	realised mde	frozen	inflow_rate	-0.076	0.086			0.242	10
2019-08	realised mde	frozen	net_rate	-0.130	0.122			0.340	10
2019-08	rank tercile	B=[V9, V6, V11, V10, V13] T=[V5, V15, V12, V7, V14]	ln_clients	+0.075	0.213	0.722			10
2019-08	rank tercile	B=[V9, V6, V11, V10, V13] T=[V5, V15, V12, V7, V14]	outflow_rate	+0.043	0.052	0.448			10
2019-08	rank tercile	B=[V9, V6, V11, V10, V13] T=[V5, V15, V12, V7, V14]	inflow_rate	-0.057	0.086	0.548			10
2019-08	rank tercile	B=[V9, V6, V11, V10, V13] T=[V5, V15, V12, V7, V14]	net_rate	-0.100	0.118	0.435			10
2019-08	bottom vs all others	G=17	ln_clients	+0.037	0.146	0.815			17
2019-08	bottom vs all others	G=17	outflow_rate	+0.018	0.039	0.659			17
2019-08	bottom vs all others	G=17	inflow_rate	-0.049	0.058	0.410			17
2019-08	bottom vs all others	G=17	net_rate	-0.067	0.083	0.454			17
2019-08	randomisation inference	frozen C(10,5)=252	ln_clients	+0.089			0.690		10
2019-08	randomisation inference	frozen C(10,5)=252	outflow_rate	+0.054			0.373		10
2019-08	randomisation inference	frozen C(10,5)=252	inflow_rate	-0.076			0.460		10
2019-08	randomisation inference	frozen C(10,5)=252	net_rate	-0.130			0.349		10
2019-08	randomisation inference	wide C(17,5)=6188	ln_clients	+0.037			0.821		17
2019-08	randomisation inference	wide C(17,5)=6188	outflow_rate	+0.018			0.691		17
2019-08	randomisation inference	wide C(17,5)=6188	inflow_rate	-0.049			0.457		17
2019-08	randomisation inference	wide C(17,5)=6188	net_rate	-0.067			0.498		17
2019-08	placebo window	fake 2018	ln_clients	-0.088	0.164	0.601			10
2019-08	placebo window	fake 2018	outflow_rate	+0.019	0.121	0.870			10
2019-08	placebo window	fake 2018	inflow_rate	-0.044	0.069	0.560			10
2019-08	placebo window	fake 2018	net_rate	-0.063	0.164	0.707			10
2019-08	control prev	ln clients frozen + ln(clients_prev)	outflow_rate	+0.028	0.064	0.677			10
2019-08	control prev	ln clients frozen + ln(clients_prev)	inflow_rate	-0.062	0.118	0.685			10
2019-08	control prev	ln clients frozen + ln(clients_prev)	net_rate	-0.090	0.163	0.616			10
2020-10	realised mde	frozen	ln_clients	-0.283	0.119			0.333	10
2020-10	realised mde	frozen	outflow_rate	-0.036	0.031			0.088	10
2020-10	realised mde	frozen	inflow_rate	-0.077	0.033			0.092	10
2020-10	realised mde	frozen	net_rate	-0.041	0.040			0.111	10
2020-10	rank tercile	B=[V2, V10, V1, V15, V7] T=[V8, V9, V6, V4, V13, V14]	ln_clients	-0.359	0.131	0.016			11
2020-10	rank tercile	B=[V2, V10, V1, V15, V7] T=[V8, V9, V6, V4, V13, V14]	outflow_rate	-0.030	0.029	0.312			11

Round	Check	Configuration	Outcome	Effect	s.e.	Boot. p	RI p	MDE	G
2020-10	rank tercile	B=[V2, V10, V1, V15, V7] T=[V8, V9, V6, V4, V13, V14]	inflow_rate	-0.157	0.081	0.050			11
2020-10	rank tercile	B=[V2, V10, V1, V15, V7] T=[V8, V9, V6, V4, V13, V14]	net_rate	-0.127	0.090	0.182			11
2020-10	bottom vs all others	G=16	ln_clients	-0.325	0.117	0.012			16
2020-10	bottom vs all others	G=16	outflow_rate	+0.001	0.036	0.983			16
2020-10	bottom vs all others	G=16	inflow_rate	-0.121	0.055	0.054			16
2020-10	bottom vs all others	G=16	net_rate	-0.122	0.068	0.101			16
2020-10	randomisation inference	frozen C(10,5)=252	ln_clients	-0.283			0.079		10
2020-10	randomisation inference	frozen C(10,5)=252	outflow_rate	-0.036			0.278		10
2020-10	randomisation inference	frozen C(10,5)=252	inflow_rate	-0.077			0.079		10
2020-10	randomisation inference	frozen C(10,5)=252	net_rate	-0.041			0.373		10
2020-10	randomisation inference	wide C(16,5)=4368	ln_clients	-0.325			0.026		16
2020-10	randomisation inference	wide C(16,5)=4368	outflow_rate	+0.001			0.992		16
2020-10	randomisation inference	wide C(16,5)=4368	inflow_rate	-0.121			0.124		16
2020-10	randomisation inference	wide C(16,5)=4368	net_rate	-0.122			0.191		16
2020-10	placebo window 2017 2019	fake 2018	ln_clients	+0.060	0.147	0.732			10
2020-10	placebo window 2017 2019	fake 2018	outflow_rate	+0.091	0.081	0.307			10
2020-10	placebo window 2017 2019	fake 2018	inflow_rate	+0.048	0.078	0.575			10
2020-10	placebo window 2017 2019	fake 2018	net_rate	-0.043	0.133	0.788			10
2020-10	control prev ln clients	frozen + ln(clients_prev)	outflow_rate	+0.016	0.046	0.713			10
2020-10	control prev ln clients	frozen + ln(clients_prev)	inflow_rate	-0.144	0.054	0.016			10
2020-10	control prev ln clients	frozen + ln(clients_prev)	net_rate	-0.160	0.083	0.038			10
2020-10	control 2019 bottom x post	frozen + 2019 bottom x post	ln_clients	-0.283	0.118	0.034			10
2020-10	control 2019 bottom x post	frozen + 2019 bottom x post	outflow_rate	-0.036	0.032	0.284			10
2020-10	control 2019 bottom x post	frozen + 2019 bottom x post	inflow_rate	-0.077	0.030	0.039			10
2020-10	control 2019 bottom x post	frozen + 2019 bottom x post	net_rate	-0.041	0.039	0.316			10

Table S36: Leave-one-verifier-out ranges

Round	Outcome	Min effect	Max effect	Min p	Max p
2019-08	inflow_rate	-0.117	-0.004	0.246	0.958
2019-08	ln_clients	-0.037	0.237	0.265	0.950
2019-08	net_rate	-0.189	-0.044	0.178	0.687
2019-08	outflow_rate	0.021	0.088	0.092	0.665
2020-10	inflow_rate	-0.094	-0.054	0.012	0.104
2020-10	ln_clients	-0.333	-0.213	0.028	0.092
2020-10	net_rate	-0.066	-0.019	0.092	0.633
2020-10	outflow_rate	-0.058	-0.020	0.044	0.493

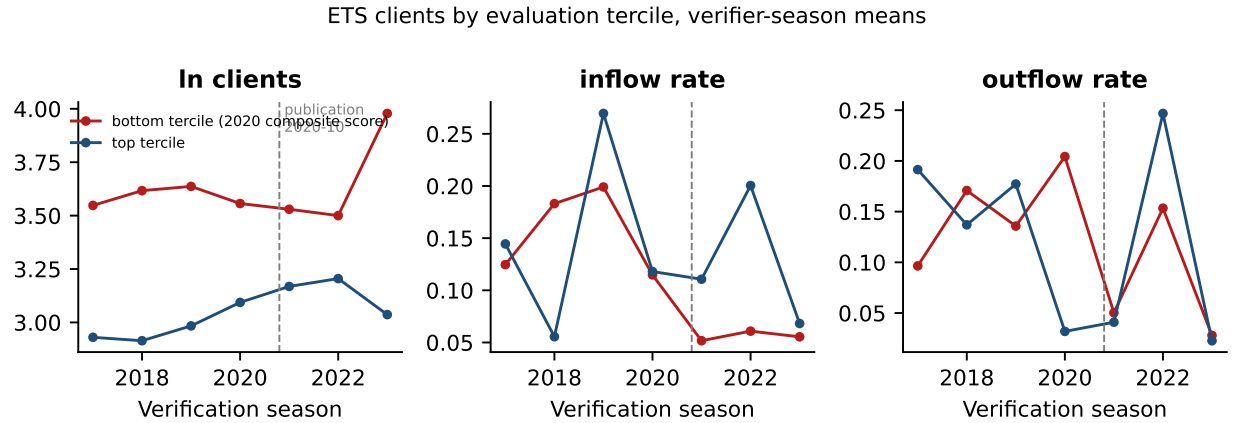


Figure S6: ETS clients by 2020 evaluation tercile: verifier-season means of ln clients, inflow rate and outflow rate.

S10. Reserved

A descriptive comparison with the market for sustainability-report assurance is left to a separate paper.

S11. Frozen analysis plan, feasibility checks and deviations

Frozen items. The analysis plan (prereg_T07.md, in Korean, in the reproduction package) was frozen on 19 September 2026 at 20:56 KST with SHA-256 hashes of the plan body and of nine input and code files. It fixed: (1) the 24-code verifier alias table; (2) 2013 as the base year, the exclusion of 2011–2012 pairs and their grey treatment in figures; (3) the switch typology (voluntary; cancellation-linked block, defined as at least half of the disappearing verifier’s last-year book moving to one successor; cancellation-linked dispersed; pilot-period dropout) and the scope-change flags; (4) the re-dating of exits to verification seasons and the classification categories from the NIER register, with ‘cause not recorded’ where the register is silent; (5) the industry split from Annex 5 (industry_split.csv, 77 labels) and the rule fixed-factor = activity class in {combustion, mobile, building, energy} and entity emissions below 500 kt; (6) the fixed-effects sample rules (energy and emissions positive, movers only, at least 30 observations per verifier, reference V1, 25 alternating-projection iterations); (7) the inference rules (small-sample cluster correction, MDE = 2.8 s.e., TOST at $\pm 2\%$ and $\pm 5\%$ on the 90% interval, wild cluster bootstrap with 999 draws and fixed seed); (8) the bound definition as the largest absolute 95% confidence limit among the seven largest verifiers, with point-estimate dispersion never used as a bound; (9) the event-study specification and the reading rule; (10) the list of at most 12 subsamples; (11) the block-transfer DiD definition; (12) the directional hypothesis for the evaluation disclosures, the first effect seasons, the placebo and the reporting of a null as ‘not detectable’; (13) the design MDE table (fixed-factor median 4.5%, discretion 19.7%, all movers 11.6%; event study 2.2–4.8 percentage points; disclosure DiD 27%); (14) a list of phrases not to be used; (15) the journal order. The feasibility checks run before the freeze (risk_checks.py) were a verifier-effect Wald test on the uncorrected covariance ($p = 0.63$; 0.84 with the small-sample correction), the raw gap in large-revision probability between switch years and other years (18.4% versus 12.6%, which fell to +1.3 percentage points, $p = 0.20$, with

entity and industry-by-year fixed effects and to +0.2, $p = 0.85$, from 2015), and a dispersion test on cancellation-linked switches (+0.9 percentage points, $p = 0.69$); none of these figures is used in the paper.

Hashes. Raw-byte SHA-256 (first 12 hexadecimal digits): verifier_panel.csv 393cbcf0f79a; switch_pairs.csv 49f0aca79143; industry_split.csv ae7bc83fe903; power_mde.csv 72b5c43f7874; external/exit_classification.csv 5632e91daa7c; external/verifier_evaluations.csv dcc48e67421d; verifier_fe.py aa291b44c943; risk_checks.py 239f650a22a4; plan_body 28e72ce96e37; wild_cluster_boot.py e50ca39e629b at the freeze and b4ca8e47dc1b after the one post-freeze change described below.

Deviations and additions logged after the freeze.

Table S37: Deviation log (summary; the full log is deviations_from_prereg.md)

Item	Content	Effect
Header wording	A pre-freeze sentence about items 4–5 remained in the frozen header; the items were complete and hashed.	none
Pre-freeze outputs	Panel, pairs, mover selection, market structure, channel decomposition and counterfactual HHI (no hypothesis tests) were produced before the freeze; feasibility-check estimates are not used.	none
Annex version	Only the current Annex 5 was obtained; earlier versions assumed identical.	stated limitation
Exit cause	The register’s remark column is blank for all six cancellations; all are described as ‘cause not recorded’.	wording
Bootstrap module	The cluster-covariance loop was vectorised after the freeze (numerically identical, maximum difference 1.4×10^{-17}); the freeze recorded a raw-byte hash and the first log entry a text-mode hash, which explains the two hash values.	none
Disclosure placebo	The frozen placebo (a fake 2017 disclosure) had no pre-period because the ETS client panel starts in season 2016; replaced by a fake 2018 disclosure within the pre-period window. A base-year bug in the placebo panel was fixed. The actual test is unchanged.	placebo only
Tercile ties	The plan had no tie rule; the frozen script broke ties by sort order. Every tie resolution and the published-rank rule are reported (S9).	sensitivity
Ladder	The raw-to-conditional ladder of S7 was recomputed on the frozen event-study sample; pre-freeze figures are not used.	descriptive
Descriptive additions	Entrants’ choices by group, the share-point decomposition, the reconciliation of the two voluntary-flow totals, the fixed-factor share of the registry and the verifier-season panel export were added (no hypotheses).	descriptive
FE robustness set	Holm adjustment, entity trends, switch-time level jumps, quiet movers, all entities, 2013–2019, cell counts, rank correlations, scope-stable bootstrap (S6).	exploratory
Disclosure robustness set	Published-rank terciles, wider control group, leave-one-out, randomisation inference, common placebo window, controls, realised MDE (S9); the entity-level LPM with leads was not run.	exploratory

S12. Reproduction package

The package contains the assembled panel (verifier_panel.csv), the consecutive-pair file (switch_pairs.csv), the industry split, the external record files (NIER register extracts, evaluation tables, exit classification, Annex 5 transcription with sources), all scripts, a driver (run_all.py) that checks the frozen

hashes, re-runs every stage and compares fifteen headline numbers with the regenerated files, the frozen analysis plan, the feasibility-check log and the deviation log. It is held in the author's repository and is available from the author on reasonable request. Python 3.12 with numpy, pandas, scipy and matplotlib is sufficient; the confirmatory stage takes about 20 minutes and the robustness stage about 40 minutes on a laptop.